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Fundamental modular region, Boltzmann factor and area law in lattice theory

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The thermodynamic limit of lattice gauge theory is derived in a gauge which is optimized to make all link variables as close to unity as possible. The derivation rests upon (1) a precise bound on the fundamental modular region and (2) a direct evaluation of the functional integral of the Wilson lattice by the saddle-point method that is valid in the thermodynamic limit. The result confirms the calculational scheme obtained previously, which differs from the Faddeev–Popov scheme by the incorporation of non-perturbative effects, but which remains perturbatively renormalizable. The lattice Faddeev–Popov propagator, which appears in the modified action, acquires a dipole singularity at zero momentum, characteristic of long range correlation. This is sufficient to produce an area law for Wilson loops, provided that unevaluated terms do not cancel the effect found.

1. Introduction

Wilson's lattice gauge theory provides a gauge-invariant regularization of quantum chromodynamics which, moreover, is well suited to numerical simulation. Such simulations, on dedicated computers, provide satisfactory agreement with the known spectrum of elementary particles, within the limitations of the systematic and statistical errors characteristic of such simulations. Nevertheless, it is not without interest to attempt to also obtain analytic results from lattice gauge theory which are complementary to the numerical results.

The lattice itself, with finite lattice spacing, is of course an artifact. The physics of hadrons is described by the critical limit of lattice gauge theory, in which a correlation length, $\Lambda_{\rm latt}^{-1}$, becomes infinite, $(a\Lambda_{\rm latt})^{-1} \to \infty$, when measured in units of the lattice spacing a. Here $\Lambda_{\rm latt}$ sets the scale for hadronic mass. The critical limit is also the weak-coupling limit, $g_0^{-2} = 2b_0 \ln[(a\Lambda_{\rm latt})^{-1}] \to \infty$, where g_0 is the

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bare coupling constant, and b_0 is a positive number. Moreover, the partition function of lattice gauge theory

$$Z = \int \mathrm{d}U \exp[-\beta S_{\mathrm{W}}(U)], \tag{1.1}$$

where S_{W} is the Wilson action and $\beta = 2ng_0^{-2}$, has the form of a (fictional) four-dimensional classical statistical mechanical lattice system for which the weak-coupling limit is the low-temperature limit. Thus we have the scheme: hadron physics = critical limit = weak-coupling limit = low-temperature limit.

Low temperature usually means small fluctuations, and a low-temperature expansion is an expansion in small fluctuations. But here the peculiarly gauge character of the theory must be considered. Because of the local gauge invariance of the Wilson action, $S_{\mathrm{W}}(U) = S_{\mathrm{W}}(U^g)$, fluctuations in gauge directions are not suppressed at all, even at low temperature. (Here U represents a lattice configuration and U^g represents its gauge transform by a local gauge transformation g_x – conventions will be given shortly.) To address this problem we minimize gauge fluctuations by choosing a gauge which makes the link variables U_l, on each link l of the lattice, as close to unity as possible in an equitable way over the whole lattice. For this purpose, we take as a measure of the deviation of the link variables from unity, the quantity

$$I(U) = \sum_l \left(1 - n^{-1} \operatorname{Re} \operatorname{tr} U_l\right) = DV - E(U), \tag{1.2a}$$

$$E(U) = \sum_l n^{-1} \operatorname{Re} \operatorname{tr} U_l, \tag{1.2b}$$

where the sum extends over all links l of the lattice. It is positive, $I(U) \geqslant 0$, and vanishes, $I(U) = 0$, if and only if $U_l = 1$ on every link l. In Appendix A, it is shown that this expression represents a (distance)^2 from U to the vacuum configuration U_0, defined by $(U_0)_l = 1$. The continuum analog of this expression is the Hilbert norm of the connection A, $I[A] = \int \mathrm{d}^D x \, |A(x)|^2$.

The restriction of this function to the gauge orbit through a fixed configuration U is given by

$$F_U(g) \equiv I(U^g). \tag{1.3}$$

Because the gauge orbit is compact and $F_U(g)$ is bounded below, at least one minimum $g = h$ exists on each gauge orbit. A gauge transformation by h brings this minimum to $g = 1$. As a preliminary gauge-fixing, we choose the set Λ of all configurations U such that this function is an absolute minimum on each gauge orbit at $g_x = 1$,

$$\Lambda \equiv \{U \colon F_U(1) \leqslant F_U(g) \text{ for all } g\}, \tag{1.4}$$

and we call Λ "the fundamental modular region". Actually, this condition can at best determine the minimum to within a global or x-independent gauge transformation, because for constant g_x we have $F_U(g) = F_U(1)$. This is not inconvenient at all, and we let it be understood that "non-degenerate absolute minimum" means "unique modulo global gauge transformations". This would be an exact gauge-fixing, and Λ would be the physical configuration space, but for the fact that, in general, there may be degenerate absolute minima,

$$F_V(h) = F_U(1) \leqslant F_U(g) \text{ for all } g,$$

with h_x a non-constant gauge transformation. In fact, in continuum gauge theory, Singer's theorem [1], inspired by Gribov's results [2], implies that for continuous connections on a compact manifold, degenerate minima do occur, and the same phenomenon presumably also occurs on a finite lattice. To this extent the gauge-fixing procedure fails. To obtain the physical configuration space P, which is the space of configurations modulo gauge transformations, the degenerate minima are identified topologically, and the resultant set Λ^* is isomorphic to the physical configuration space P,

$$\Lambda^* = \mathrm{P} = U/\mathrm{G}. \tag{1.5}$$

How serious is the problem of degenerate absolute minima? In continuum gauge theory, it is known for a class of connections that Λ is convex [3], and that its interior consists of non-degenerate absolute minima [4]. Degenerate absolute minima can, and do, occur only on the boundary $\partial\Lambda$. In other words, the interior of Λ is free of Gribov copies, but different points on the boundary may be Gribov copies of each other. For a finite lattice, partial results were obtained in ref. [5]. In Appendix A of the present article, it is proven for a finite lattice that the interior of Λ consists of non-degenerate absolute minima, so Gribov copies may only occur on the boundary $\partial\Lambda$. Consequently they may be ignored, because the boundary $\partial\Lambda$ where degenerate minima may occur, has measure zero in the formula for the partition function of a finite lattice, eq. (1.12) below.

(Conventions are as follows. The lattice is hypercubic, D-dimensional, periodic, with edge of integer length L, and volume $V = L^D$. The lattice spacing a is chosen as the unit of length. Sites are labelled by D-dimensional vectors x^μ, $\mu = 1, \ldots, D$, with integer components, $x^\mu = 1, \ldots, L$. The link variable $U_\ell \in \mathrm{SU}(n)$ associated to the link ℓ from x to y is denoted U_{xy}, with $U_{yx} = (U_{xy})^{-1} = U_{xy}^\dagger$, or alternatively, on the link from x to $x + e_\mu$, by $U_\mu(x)$, where e_μ is a unit vector in the positive μ-direction, with $U_\mu(x) = U_\mu(x + Le_\mu)$. The gauge transform of the link variable U_{xy} is written $(U^g)_{xy} = g_x^\dagger U_{xy} g_y$, where g_x and g_y are elements of $\mathrm{SU}(n)$. A basis of the $\mathrm{SU}(n)$ Lie algebra in the fundamental representation is provided by traceless, anti-hermitian $n \times n$ matrices t^a, normalized to $\mathrm{tr}(t^a t^b) = -\frac{1}{2}\delta^{ab}$, which satisfy $[t^a, t^b] = f^{abc} t^c$. We adopt the shorthand custom of writing the partition function Z, to also represent the generic expectation value of a gauge-invariant quantity.)

The configurations in the fundamental modular region (1.4) are most easily described in terms of the variables $A_\mu^a(x)$ which are the traceless anti-hermitian part of $U_\mu(x)$,

$$A_\mu^a(x) t^a = 2^{-1}\left[U_\mu(x) - U_\mu^\dagger(x)\right]_{\text{traceless}}. \tag{1.6}$$

These variables approach the classical gauge connection in the continuum limit. At any local or absolute minimum of the minimizing function, its first variation vanishes and the matrix of its second derivatives is non-negative. As shown in Appendix B, these two conditions, at $g_x = 1$, are given respectively by the transversality of $A_\mu^a(x)$,

$$\nabla \cdot A = 0, \tag{1.7}$$

where $\nabla \cdot A$ is the lattice divergence of A, given in eq. (B.6), and the positivity of the lattice Faddeev–Popov matrix $M(U)$, given in eq. (B.18),

$$M(U) \geqslant 0. \tag{1.8}$$

The first condition defines the gauge-fixing surface Γ,

$$\Gamma \equiv \{U\colon \nabla \cdot A(U) = 0\}, \tag{1.9}$$

so the gauge just defined falls into the class of lattice Landau gauges, and we call it the "minimal Landau gauge". The two conditions together define the Gribov region Ω,

$$\Omega \equiv \{U\colon \nabla \cdot A(U) = 0 \text{ and } M(U) \geqslant 0\}. \tag{1.10}$$

Because the set Λ of absolute minima is contained in the set of relative minima, we have the inclusion

$$\Lambda \subset \Omega, \tag{1.11}$$

where Λ is the fundamental modular region.

The partition function, or measure on physical configurations on a lattice of edge L, is the integral over the fundamental modular region Λ_L, weighted by the Faddeev–Popov determinant,

$$Z(\Lambda_L) \equiv \int_{\Lambda_L} \mathrm{d}U \exp[-\beta S_{\mathrm{W}}(U)]\, \delta(\nabla \cdot A) \det[M(U)]. \tag{1.12}$$

The concrete problem which this article addresses is to evaluate this integral at large volume V, with the cutoff at the boundary $\partial \Lambda$ in place, and in the two concluding sections to derive non-perturbative results.

In ref. [5] this expression was evaluated in the critical limit (infinite-volume *and* weak-coupling limit) using earlier results [6,7]. The result is expressed as a calculational scheme which differs from the Faddeev–Popov scheme by the inclusion of non-perturbative effects, but which remains perturbatively renormalizable. However, the derivation in ref. [5] relied to a considerable extent on physical intuition and a formal weak-coupling expansion. In the present article, we evaluate formula (1.12) in the thermodynamic limit (infinite-volume limit only) from first principles, by a new method that does not rely on a weak-coupling expansion. The previous result is confirmed in the weak-coupling limit. The new derivation does however require two hypotheses which are mentioned in the following paragraphs (see also eq. (2.6) for hypothesis (i) and the discussion preceding eq. (7.12) for hypothesis (ii)).

The contents and results of this article are as follows. The basic idea of the present approach is developed in sect. 2. It consists in taking the infinite-volume limit in two steps, first for the gauge transformations and then for configurations, as we now explain. In step one, periodicity restrictions on local gauge transformations g are relaxed so that they are defined on an infinite lattice, whereas the configurations U are required to have period L. With this enlargement of the class of allowed gauge transformations, the condition (1.4) on configurations U, of being an absolute minimum with respect to all gauge transformations in the class, becomes stricter. This stricter condition defines "the core region" Ξ_L, a set of configurations which is smaller than the fundamental modular region Λ_L, but contained in it, $\Xi_L \subset \Lambda_L$. A partition function associated with the core region $Z(\Xi_L)$ is defined by replacing the fundamental modular region Λ_L by the core region Ξ_L in formula (1.12) for the partition function. In step two, periodicity conditions on configurations are relaxed, namely the infinite-volume limit is taken in the form $\lim_{L\to\infty} Z(\Xi_L)$. It seems reasonable that the desired infinite-volume limit should be obtained in this way, because relaxation of the periodicity condition on gauge transformations at finite volume should not be important after the infinite-volume limit on configurations is taken in the second step. This constitutes hypothesis (i) of the present article, namely, $\lim_{L\to\infty} Z(\Xi_L) = \lim_{L\to\infty} Z(\Lambda_L)$. The advantage of this procedure is that it is simpler to locate the boundary of Ξ_L than Λ_L. In sect. 2, we obtain an explicit formula, given in theorem 1, for the minimizing gauge transformation g_x on an infinite lattice for configurations of period L. It closely resembles the Bloch wave function, even though the minimization problem is non-linear.

In sect. 3, we obtain explicit bounds on the core region Ξ_L, described in theorem 2, which are saturated on part of its boundary $\partial\Xi_L$ which we call the "distinguished boundary". In sect. 4, simple bounds on Ξ_L are derived, which are expressed in terms of the "horizon function" $H(U)$, which is defined by

$$H(U) \equiv \sum_{x,y} \left[D^{ab}_{x,\mu}(U) D^{ac}_{y,\mu}(U) M^{-1}_{x,b;y,c}(U) \right]$$

$$-(n^2-1)E(U). \tag{1.13}$$

Here repeated indices μ, a, b and c are summed over, $E(U)$ is the maximizing function, eq. (1.2b), $D^{ab}_{x,\mu}(U)$ is the lattice gauge-covariant derivative defined in eq. (B.11), and $M^{-1}_{x,b;y,c}(U)$ is the inverse of the lattice Faddeev–Popov matrix M. In particular, according to theorem 3 of sect. 4, configurations U in the core region Ξ_L satisfy the bound $H(U) \leqslant 0$. As the boundary $\partial\Omega_L$ of the Gribov region Ω_L is approached from the interior, the horizon function diverges in general, $H(U) \to \infty$, because $M(U)$ develops a vanishing eigenvalue there. In sect. 4 "the augmented core region" Ψ_L, is introduced, which has the simple definition, $H(U) \leqslant 0$, and satisfies $\Xi_L \subset \Psi_L \subset \Omega_L$.

In sect. 5, the partition function associated with the augmented core $Z(\Psi_L)$ is evaluated in the infinite-volume limit by the saddle-point method, with the result that, for the purpose of taking the infinite-volume limit, the cutoff at $\partial\Psi_L$ may be replaced by the Boltzmann factor $\exp[-\alpha H(U)]$. More precisely, as stated in theorem 4, let the partition function $Z(\alpha)$, which contains this Boltzmann factor and a cutoff at the boundary of the Gribov region $\partial\Omega_L$, be defined by

$$Z(\alpha) \equiv \int_{\Omega_L} \mathrm{d}U \exp[-\beta S_{\mathrm{W}}(U) - \alpha H(U)] \delta(\nabla \cdot A) \det[M(U)]. \tag{1.14}$$

Then we have $\lim_{L\to\infty} Z(\Psi_L) = \lim_{L\to\infty} Z(\alpha_{\mathrm{cr}})$, where the value α_{cr} of the thermodynamic parameter α, is a critical value determined by the "horizon condition",

$$\langle H \rangle = -\alpha^{-1}, \tag{1.15}$$

and the expectation value is calculated in the ensemble $Z(\alpha)$. The Boltzmann factor $\exp[-\alpha H(U)]$ has an essential singularity at the boundary $\partial\Omega_L$ of the Gribov region, and vanishes exponentially there.

In sect. 6, auxiliary or ghost fields are introduced which allow $Z(\alpha)$ to be expressed in terms of a local lattice action. From it, a local weak-coupling perturbative expansion of $Z(\alpha)$ may be derived, which is insensitive to the cutoff at $\partial\Omega_L$ in eq. (1.14). This allows a calculational scheme in two steps. In step (i) eq. (1.14) is evaluated perturbatively, and in step (ii) eq. (1.15) is solved non-perturbatively. At the non-perturbative level, the theory in functional-integral form remains non-local, because of the restriction to the Gribov region Ω_L, however the non-locality will be eliminated non-perturbatively in sect. 9 in the Schwinger–Dyson formulation.

In sect. 7, it is shown (theorem 5) that in the infinite-volume limit, the probability distribution defined by $Z(\alpha)$ for $\alpha = \alpha_{cr}$, which satisfies $Z(\Psi) = Z(\alpha_{cr})$, gets concentrated on a submanifold of the configuration space where certain second and third rank tensors vanish. These conditions hold on a part of the boundary of the core region Ξ, which is a subset of the distinguished boundary, and which we call "the extreme boundary". Hypothesis (ii) of the present article is that these conditions are also sufficient for a configuration in the Gribov region Ω to lie on the extreme boundary of the core region Ξ. We take this as a working hypothesis, whose correctness is to be judged by the consistency of the results. This completes the derivation of the Boltzmann formula, for with this hypothesis it follows that $Z(\Psi)$ vanishes on the difference set $\Delta = \Psi - \Xi$ in the infinite-volume limit, and we obtain $Z(\Lambda) = Z(\Xi) = Z(\Psi) = Z(\alpha_{cr})$.

It was shown recently [8] that the naive continuum limit of the local lattice action, derived in sect. 6, is perturbatively renormalizable. This was done by showing that the action is BRS invariant when α is set to zero, and by controlling the BRS-violating term which is proportional to α, by introducing a local source for it and its BRS transform. The basic reason for this is that α has dimensions $(\text{mass})^4$, so the BRS-breaking term is extremely soft. In sect. 8 of the present article it is shown that the local lattice action, derived in sect. 6, is BRS invariant when $\alpha = 0$, and consequently renormalizability follows also from the lattice regularization. (The remaining sections do not depend on sects. 8 and 9 or their dependent Appendices G and H, which may be skipped in a first reading.)

In sect. 9, an effective action is defined for lattice gauge theory. Because there exists no global coordinatization on the group manifold, this is done by writing the partition function as an integral over the linear span of the group manifold, and by expressing the non-linear conditions which define a matrix group, such as $u^{\dagger}u = 1$, by means of Lagrange-multiplier fields. Local sources are then introduced for all the linearly independent fields and their BRS transforms, including the Lagrange multipliers. In this way, the Haar measure becomes cartesian and the lattice action becomes polynomial. This allows us to write non-perturbative Schwinger–Dyson equations that are local.

A systematic lattice perturbation theory based on this approach, which does not rely on the exponential mapping to represent group elements, is outlined in Appendix H.

In sect. 10, as a first non-perturbative application, it is shown that the horizon condition is equivalent to the condition that the lattice Faddeev–Popov propagator $G(\theta)$ has a dipole singularity at zero momentum, $G^{-1}(\theta) \sim c\lambda^2(\theta)$, which corresponds to particularly long range correlations. Here $G(\theta)$ is the Fourier transform of $\langle M_{x,y}^{-1} \rangle$, where $M_{x,y}$ is the lattice Faddeev–Popov matrix, θ_{μ} is the Brillouin momentum, and $\lambda(\theta) \equiv \sum_{\mu}[4 \sin^2(\frac{1}{2}\theta_{\mu})]$. The continuum analog of this result was proven in ref. [8]. It is a highly non-perturbative result, for it is equivalent to the statement that the quantum contributions to the Faddeev–Popov self-energy precisely cancel the zero-order self-energy $\lambda(\theta)$ at $\theta = 0$.

Because M^{-1} appears in the action αH, eq. (1.13), it is expected that other correlation functions also change qualitatively in the limit $\alpha \to \alpha_{cr}$, including in particular the gluon propagator $D(\theta)$ and the Wilson loop W. The behavior of the gluon propagator $D(\theta)$ at zero momentum is discussed at the end of sect. 10.

In the concluding sect. 11, the effect of the dipole singularity in $\langle M^{-1} \rangle$ on the expectation value W of a Wilson loop corresponding to a closed planar curve surrounding a large area A is calculated. For this purpose, the horizon condition is solved in the form $g = g_{cr}(\alpha)$, so α is the independent parameter instead of g, as is appropriate for the continuum limit, because, after renormalization, $\alpha \sim \Lambda_{latt}^4$. A non-perturbative evaluation of the simplest term gives

$$\alpha^{1/2} \partial W / \partial \alpha^{1/2} = -c \alpha^{1/2} A W + \text{remainder}, \tag{1.16}$$

where $c > 0$ is dimensionless, $\alpha^{1/2}$ has the dimension of string tension, and the first term is obtained by evaluating the simplest non-perturbative contribution. For an area law to be established, it is sufficient that the remainder not completely cancel this result. To estimate the remainder reliably is a challenging problem for future investigation.

2. The core of the fundamental modular region

A source of considerable difficulty in determining the absolute minima of the minimizing function $F_U(g)$ on a hypercubic periodic lattice of edge L is the constraint that the local gauge transformations $g(x)$ have period L. On the other hand, periodicity is unimportant in the infinite-volume limit $L \to \infty$, which is our only concern. Consequently we shall simplify the minimization problem by relaxing the constraint on $g(x)$ as follows.

Given the euclidean hypercubic periodic lattice of edge L on which the link variables $U_\mu(x)$ are defined, we consider a larger periodic hypercubic lattice, of edge $P = NL$, which we regard as being constituted of N^D blocks of the original lattice. Every configuration U on the hypercubic lattice of edge L may be identified in a natural way with a unique configuration on the larger lattice of edge NL, and we denote corresponding configurations on the two lattices by the same symbol, and similarly for sets of such configurations. A new minimization problem is defined by minimizing with respect to all local gauge transformations $g(x)$ that have period NL, but the configurations U are, as before, required to have period L. Call the corresponding set of absolute minima defined in this way,

$$\Lambda_L{}^N \equiv \{U\colon U \in \Pi_L \text{ and } F_U(1) \leqslant F_U(g) \text{ for all } g \in \mathrm{G}_{NL}\}, \tag{2.1}$$

"the Nth partial core of the fundamental modular region". Here Π_L is the set of configurations with period L, and G_{NL} is the set of local gauge transformations with period NL. We may also write

$$\Lambda_L{}^N = \Lambda_{LN} \cap \Pi_L. \tag{2.2}$$

All gauge transformations of period L are included in those of period NL, so we have the inclusion

$$\Lambda_L{}^N \subset \Lambda_L. \tag{2.3}$$

For the same reason, if N is chosen to be a power of 2, say, then the partial cores are nested,

$$\Lambda_L{}^{N'} \subset \Lambda_L{}^N \subset \Lambda_L \quad \text{for } N' = 2^{M'} > N = 2^M. \tag{2.4}$$

Call the limiting set,

$$\Xi_L \equiv \Lambda_L^\infty = \lim_{N \to \infty} \Lambda_L{}^N, \tag{2.5}$$

the "minimal core of the fundamental modular region", or simply "the core region". We shall see shortly (remark 4, below) that Ξ_L is independent of the sequence N.

We suppose (hypothesis (i)) that all trace of periodicity is lost in the infinite-volume limit, $L \to \infty$, in the sense that the core region and the fundamental modular region give the same limiting correlation functions,

$$\lim_{L \to \infty} Z(\Xi_L) = \lim_{L \to \infty} Z(\Lambda_L), \tag{2.6}$$

where $Z(\Lambda_L)$ is defined in eq. (1.12), and $Z(\Xi_L)$ is similarly defined, so it is sufficient to determine the core region Ξ_L instead of the fundamental modular region Λ_L.

Determination of the core region is greatly facilitated by the following theorem, proven in Appendix F, which gives the form of the minimizing gauge transformation.

Theorem 1. Let $U_\mu(x)$ be a configuration of period L, $U_\mu(x + Le_\nu) = U_\mu(x)$, and let $g(x)$ be the minimizing gauge transformation on the lattice of edge NL, so that U^g is an interior point of Λ_{NL}. Then $g(x)$ is of the form

$$g(x) = h(x) \exp\left(t^a \theta_\mu^a x^\mu\right), \tag{2.7}$$

where $h(x)$ has period L, $h(x + Le_\mu) = h(x)$. The θ_μ^a are elements of a Cartan subalgebra,

$$\left[t^b \theta_\mu^b, t^c \theta_\nu^c\right] = t^a f^{abc} \theta_\mu^b \theta_\nu^c = 0, \tag{2.8}$$

and for each μ have eigenvalues restricted by

$$t^a\theta^a_\mu\psi^i = \left(2\pi i n^i_\mu/NL\right)\psi^i, \tag{2.9}$$

where the n^i_μ are integers. The quantities $h(x)$ and θ^a_μ are further restricted by the condition

$$2^{-1}\sum_\mu\left\{\left[U^h_\mu(x)\ \exp\left(t^a\theta^a_\mu\right) - \exp\left(t^a\theta^a_\mu\right)U^h_\mu(x-e_\mu) - \text{h.c.}\right]\right\}_{\text{traceless}} = 0. \tag{2.10a}$$

where $U^h_\mu(x) \equiv h^\dagger(x)U_\mu(x)h(x+e_\mu)$, and the subscript means the traceless part is taken. Summation of the last equation over all x gives

$$\left[\sum_{x,\mu} U^h_\mu(x),\ \exp\left(t^a\theta^a_\mu\right)\right] - \text{h.c.} = 0. \tag{2.10b}$$

Remark 1. Although the minimizing gauge transformation $g(x)$ is an element of a non-linear space which is the solution to the non-linear problem of finding an absolute minimum, nevertheless $g(x)$ has the simple Bloch wave form familiar in linear problems, generalized so that the Bloch momenta $t^a\theta^a_\mu$ takes values in a Cartan subalgebra.

Remark 2. Eq. (2.10) expresses the fact that the minimizing configuration U^g is transverse. Because the form of the gauge transformation $g(x) = h(x)\ \exp(t^a\theta^a_\mu x^\mu)$, given in eqs. (2.7)–(2.10), is compatible with transversality, it is a useful ansatz for Gribov copies that are not absolute minima, but which may be relative minima or saddle points of the minimizing function. It is also a valuable ansatz for Gribov copies obtained by gauge transformations $g(x)$ of period L. These are of interest because they determine the boundary of the fundamental modular region Λ_L. To ensure that the gauge transformation $g(x)$ has period L we require that the eigenvalues of $t^a\theta^a_\mu$ be given by

$$t^a\theta^a_\mu\psi^i = \left(2\pi i n^i_\mu/L\right)\psi^i, \tag{2.11a}$$

which is a special case of eq. (2.9). The ansatz is completed by stipulating that $h(x)$ have a generator of period L,

$$h(x) = \exp[\omega(x)], \quad \omega(x+Le_\mu) = \omega(x). \tag{2.11b}$$

As an example, consider Gribov copies of the vacuum $U_\mu(x) = 1$. This configuration is invariant under all translations, and consequently $h(x)$ is an x-independent gauge transformation, so without loss of generality we may choose $h(x) = 1$. This gives

$$g(x) = \exp\left(t^a \theta_\mu^a x^\mu\right),$$

and the Gribov copies of the vacuum are given by

$$U_\mu^g(x) = \exp\left(t^a \theta_\mu^a\right).$$

Being constant, these are indeed transverse configurations, as desired. Note that with the eigenvalues (2.11a), the gauge transformation $\exp(t^a \theta_\mu^a x^\mu)$ is "large" because for some x it is very far from unity, even though the gauge transform $U_\mu^g(x) = \exp(t^a \theta_\mu^a)$ is very close to unity when L is large and the n_μ^i are small integers. This example shows that "large" gauge transformations should not be ignored even if one were only interested in developing a (credible) perturbation theory, because the infinite-volume limit is taken before the weak-coupling limit.

Remark 3. In the case of the SU(2) group, the Cartan subalgebra is one-dimensional so $t^a \theta_\mu^a$ given in eqs. (2.8) and (2.9) is necessarily of the form

$$t^a \theta_\mu^a = \left(2\pi n_\mu / NL\right) 2t \cdot v \equiv \theta_\mu 2t \cdot v, \tag{2.12}$$

where v^a is a real unit color vector, $v \cdot v = 1$, so $(2t \cdot v)^2 = -1$. We may write $2t \cdot v = iu\sigma_3 u^\dagger$, where $u \in \mathrm{SU}(2)$, and σ_3 is the diagonal Pauli matrix. All expressions simplify because

$$\exp\left(t^a \theta_\mu^a\right) = \cos\theta_\mu + 2t \cdot v \sin\theta_\mu. \tag{2.13}$$

In particular, eq. (2.10) simplifies to

$$f^{abc}\left(\sum_{x,\mu} \sin\theta_\mu (A^h)_\mu^b(x)\right) v^c = 0, \tag{2.14}$$

where $(A^h) \equiv A(U^h)$. For the SU(3) group which is of second rank we may write

$$t^a \theta_\mu^a = \theta_{3\mu} \cdot 2t \cdot v_3 + \theta_{8\mu} \cdot 2t \cdot v_8 = u\left(i\theta_{3\mu}\lambda_3 + i\theta_{8\mu}\lambda_8\right)u^\dagger, \tag{2.15}$$

where λ_3 and λ_8 are diagonal Gell-Mann matrices, and $u \in \mathrm{SU}(3)$. This gives

$$\exp\left(t^a \theta_\mu^a\right) = c_{0\mu} I + c_{3\mu} 2t \cdot v_3 + c_{8\mu} 2t \cdot v_8, \tag{2.16}$$

where

$$c_{0\mu} = 3^{-1}\left[2\exp\left(i\theta_{8\mu}/\sqrt{3}\right)\cos\theta_{3\mu} + \exp\left(-2i\theta_{8\mu}/\sqrt{3}\right)\right],$$

$$c_{3\mu} = \exp\left(i\theta_{8\mu}/\sqrt{3}\right)\sin\theta_{3\mu},$$

$$c_{8\mu} = -i\left(\sqrt{3}\right)^{-1}\left[\exp\left(i\theta_{8\mu}/\sqrt{3}\right)\cos\theta_{3\mu} - \exp\left(-2i\theta_{8\mu}/\sqrt{3}\right)\right].$$

Remark 4. Because of the cyclicity of the trace, the minimizing function $I(U^g)$, eq. (1.3), for a lattice of edge NL and g of the form (2.7) is given by

$$I(U^g) = J(U^h, \theta), \tag{2.17}$$

where

$$J(U, \theta) \equiv \sum_{x,\mu}\left\{1 - n^{-1}\operatorname{Re}\operatorname{tr}\left[U_\mu^h(x)\exp\left(t^a\theta_\mu^a\right)\right]\right\}. \tag{2.18}$$

The argument has period L, and consequently it is sufficient to sum over the volume $V = L^D$ of the lattice of period L. We may regard $J(U^h, \theta)$ as a new minimizing function defined on the lattice of period L, and the core region Ξ_L may be defined as the set of U such that $J(U^h, \theta)$ is an absolute minimum at $h = 1$ and $\theta = 0$ with respect to all gauge transformations $h(x)$ of period L, and all θ_μ^a in a Cartan subalgebra C,

$$\Xi_L = \left\{U\colon I(U) \leqslant J(U, \theta) \text{ for all } h(x) \text{ and } \theta_\mu^a \in \mathrm{C}\right\}. \tag{2.19}$$

This shows that the core region $\Xi_L = \Lambda_L^\infty$ is independent of the sequence N of ${\Lambda_L}^{\mathrm{N}}$ used to define it. To find the boundary of Ξ_L it is sufficient to compare $I(U)$ with $J(U^h, \theta)$ for $h(x)$ and θ chosen of minimizing form. It is simple to verify that the transversality condition (2.10) is the statement that $J(U^h, 0)$ is stationary with respect to infinitesimal variations of $h(x)$ at a minimum. At the minimum, $J(U^h, \theta)$ is also invariant under infinitesimal variations of the θ which preserve the relations of a Cartan subalgebra, namely, with $\theta_\mu^a \to \theta_\mu^a + \varepsilon\eta_\mu^a$, then, to first order in ε,

$$f^{abc}\left(\theta_\mu^b\eta_\nu^c + \eta_\mu^b\theta_\nu^c\right) = 0. \tag{2.20}$$

This gives the condition

$$\sum_{x,\mu}\operatorname{Re}\operatorname{tr}\left[U_\mu^h(x)\exp\left(t^a\theta_\mu^a\right)C_{ab}(\theta_\mu)\eta_\mu^a t^b\right] = 0, \tag{2.21}$$

where η_μ^a satisfies eq. (2.20) and $C_{ab}(\alpha)$ is the Maurer–Cartan form

$$C_{ab}(\alpha)t^b \equiv \exp(-t^c\alpha^c)\frac{\partial \exp(t^c\alpha^c)}{\partial\alpha^a}.$$

Remark 5. All configurations U in Ξ_L have the property that the traceless anti-hermitian part of U, namely, $A_\mu^a(x)t^a = 2^{-1}[U_\mu(x) - U_\mu^\dagger(x)]_{\text{traceless}}$, has vanishing zero-frequency component,

$$\sum_x A_\mu^a(x) = 0. \tag{2.22}$$

To see this, choose $h(x) = 1$, so that by eq. (2.19) we have, for all U in Ξ_L and all θ_μ^a in a Cartan subalgebra,

$$\sum_{x,\mu} \text{Re tr}\big[U_m(x) - U_\mu(x)\exp(t^a\theta_\mu^a)\big] \geqslant 0,$$

With $\theta_\mu^a = \epsilon\eta_\mu^a$ and ε sufficiently small, the left hand side has the sign of the term linear in ε, so we must have

$$-\sum_{x,\mu} \text{Re tr}\big[U_\mu(x)t^a\big]\eta_\mu^a$$
$$= -2^{-1}\sum_{x,\mu}\text{tr}\Big\{\big[U_\mu(x) - U_\mu^\dagger(x)\big]t^a\Big\}\eta_\mu^a = 0,$$

where the anti-hermiticity of the t^a has been used. This is true for all η_μ^a in every Cartan subalgebra. These span the Lorentz–color vector space, and the assertion follows. (We note parenthetically that eq. (2.22) and the transversality of A allow us to write for a finite lattice, and for all U in Ξ_L, in Cartan notation, $A = {}^*d^*\Pi$, where Π is a lattice Hertz potential defined on plaquettes.) Eq. (2.22) implies that Ξ_L has $(n^2-1)D$ fewer dimensions that Λ_L. By convention we shall use the term "boundary of Ξ_L" to mean the relative boundary within the hyperplane defined by eq. (2.22) rather than all of Ξ_L. It is convenient to introduce the restricted fundamental modular region $\Lambda_{L,\text{r}}$ and the restricted Gribov region $\Omega_{L,\text{r}}$, and the restricted transverse hyperplane $\Gamma_{L,\text{r}}$ which is defined to be the subset of these regions with vanishing zero-frequency component,

$$\Lambda_{L,\text{r}} \equiv \Lambda_L \cap \Big\{U\colon \textstyle\sum_x A(U)_\mu^b(x) = 0\Big\},$$

$$\Omega_{L,\text{r}} \equiv \Omega_L \cap \Big\{U\colon \textstyle\sum_x A(U)_\mu^b(x) = 0\Big\},$$

$$\Gamma_{L,\text{r}} \equiv \Gamma_L \cap \Big\{U\colon \textstyle\sum_x A(U)_\mu^b(x) = 0\Big\}, \tag{2.23a}$$

and we have

$$\Xi_L \subset \Lambda_{L,\mathrm{r}} \subset \Omega_{L,\mathrm{r}} \subset \Gamma_{L,\mathrm{r}}. \tag{2.23b}$$

We next characterize parts of the boundary $\partial \Xi_L = \partial \Lambda_L^\infty$ of the core region, according to the change in the minimizing gauge transformation $g(x)$ that occurs there. For $N = \infty$, the restriction (2.9) on the eigenvalues of the $t^b \theta_\mu^b$ becomes vacuous, and the θ_μ^b become continuous parameters subject only to the restriction that they lie in a Cartan subalgebra, $f^{abc}\theta_\mu^b \theta_\nu^c = 0$. Now consider a part of the boundary of the core $\partial \Xi_L$ that lies in the interior of Λ_L. Along any curve which crosses this part of the boundary, the minimizing gauge transformation $g(x)$ changes from $g(x) = 1$, to $g(x) = h(x)\exp(t^a \theta_\mu^a x^\mu)$. There are two possibilities: (1) As the boundary is crossed, the parameters θ_μ^a and the gauge transformation $h(x)$ both evolve continuously from $\theta_\mu^a = 0$ and $h(x) = 1$. In other words, the minimizing gauge transformation develops from the low-frequency end of the first Brillouin zone. (2) Both θ_μ^a and $h(x)$ suffer a discontinuous jump. We call that part of the boundary Ξ_L where $g(x)$ changes according to case (1) "the distinguished boundary". It will be of interest to us because the probability weight gets concentrated on a subset of the distinguished boundary in the infinite-L limit. Note that although $h(x)$ and θ_μ evolve continuously from $h(x) = 1$ and $\theta_\mu = 0$ when the distinguished boundary is crossed, nevertheless, the gauge transformation $g(x)$ is "large" for arbitrarily small eigenvalues $2\pi n_\mu / NL$ of θ_μ because, with x^μ in the interval $1 \leqslant x^\mu \leqslant NL$, $g(x)$ is very far from unity for some x.

Properties of the boundary of the core region Ξ_L are illuminated by simple examples for which exact calculations are possible. In Appendix C it is shown that the configuration

$$\begin{aligned} U_1(x) &= \cos\beta + 2t^3(-1)^{x_2}\sin\beta, \\ U_2(x) &= \cos\beta + 2t^3(-1)^{x_1}\sin\beta, \\ U_\mu(x) &= 1 \quad \text{for } \mu > 2, \end{aligned} \tag{2.24}$$

lies inside the core region for $\beta < \frac{1}{4}\pi$, and on the distinguished boundary for $\beta = \frac{1}{4}\pi$. Thus the distinguished boundary is not empty. In Appendix D, the complete Brillouin zone of Gribov copies is found explicitly for the class of configurations of the form

$$\begin{aligned} U_\mu(x) &= 1 \quad \text{for } \mu \neq 2, \\ U_2(x) &= \cos\beta + 2t^a u^a(x^1)\sin\beta, \end{aligned} \tag{2.25}$$

where

$$t^a u^a(x^1) \equiv t^1 \cos(2\pi m x^1/L) + t^2 \sin(2\pi m x^1/L),$$

and m is an arbitrary integer. As shown in Appendix D, for configurations of this form to lie inside the core region Ξ_L, the parameter β satisfies the bound

$$\sin^2\beta/\cos\beta \leqslant \sin^2(\pi m/L) \quad \left(\text{with} |\beta| < \tfrac{1}{2}\pi\right). \tag{2.26}$$

This example shows the proximity of the boundary to the vacuum configuration, $U_\mu(x) = 1$, in infrared directions that is, for slowly varying configurations (small m/L). The proximity of the horizon in infrared directions was already observed in continuum theory by Gribov [2], and is confirmed by a rigorous ellipsoidal bound in lattice gauge theory, in Appendix B of ref. [6]. In Appendix E, the configuration

$$\begin{aligned} U_1(x) &= \cos\beta + 2t_3(-1)^{x_1+x_2}\sin\beta, \\ U_2(x) &= \cos\beta - 2t_3(-1)^{x_1+x_2}\sin\beta, \\ U_\mu(x) &= 1 \quad \text{for } \mu > 2, \end{aligned} \tag{2.27}$$

is considered. When $\beta = \frac{1}{4}\pi$, this is an example of extreme local curvature, with $U_p = -1$, for every plaquette p in the 1–2 plane. It is shown that this configuration lies inside the core region for $\beta < \frac{1}{4}\pi$, and on a degenerate gauge orbit which touches the distinguished boundary of the core region for $\beta = \frac{1}{4}\pi$.

All examples of configurations on the boundary of the core region lie in fact on the distinguished boundary. This is quite remarkable, because examples (2.25) and (2.27) have the highest possible lattice frequency, and example (2.27) is a worst case scenario, in the sense that the local curvature is a maximum. Thus it is possible that the distinguished boundary is the complete boundary of the core region.

3. Location of the distinguished boundary

In this section we shall derive equations, given in theorem 2 below, which locate the distinguished boundary of the core region Ξ_L. Recall that when the boundary is crossed, the minimizing gauge transformation changes from $g(x) = 1$ to $g(x) = h(x)\exp(t^a\theta^a_\mu x^\mu)$, as given in eq. (2.7), and that on the distinguished boundary, the Bloch momenta θ^a_μ and the gauge transformation $h(x)$ of period L develop continuously from $\theta^a_\mu = 0$ and $h(x) = 1$ respectively. In order to find the distinguished boundary we seek a minimizing gauge transformation of the form

$$\theta^a_\mu = \varepsilon\eta^a_{(1)\mu} + \varepsilon^2\eta^a_{(2)\mu} + \varepsilon^3\eta^a_{(3)\mu} + \dots, \tag{3.1}$$

$$h(x) = \exp\left[\varepsilon\omega_{(1)}(x) + \varepsilon^2\omega_{(2)}(x) + \varepsilon^3\omega_{(3)}(x) + \dots\right]. \tag{3.2}$$

Here ε is a measure of the magnitude of the Brillouin momenta θ^a_μ, and $\eta^a_{(1)\mu}$ is normalized to unity and lies in a Cartan subalgebra,

$$|\eta_{(1)}|^2 = \sum_{\mu,a} \left[\eta^a_{(1)\mu}\right]^2 = 1, \qquad f^{abc}\eta^a_{(1)\mu}\eta^b_{(1)\nu} = 0. \tag{3.3}$$

Lemma 1. Let U be a configuration in the restricted Gribov region $\Omega_{L,\mathrm{r}}$. Then for any $\eta^a_{(1)\mu}$ satisfying conditions (3.3), the transversality condition (2.10) is satisfied to first order in ε, provided that $\omega^a_{(1)}(x)$ is given by

$$\omega^a_{(1)}(x) = -\eta^b_{(1)\mu}\sum_y D^{bc}_{y,\mu}(U)M^{-1}_{x,a;y,c}(U), \tag{3.4}$$

where $D^{bc}_{y,\mu}(U)$ is the lattice gauge-covariant derivative, and $M(U)$ is the lattice Faddeev–Popov matrix $M(U) = -\nabla\cdot D(U)$, which are given in eqs. (B.11) and eq. (B.18). We write this as

$$\omega^a_{(1)}(x) = \sum_y M(U)^{-1}_{x,a;y,c}B^c_{\mu b}(y)\eta^b_{(1)\mu} \tag{3.5}$$

or simply

$$\omega^a_{(1)}(x) = \left\{M(U)^{-1}\left[B\cdot\eta_{(1)}\right]\right\}^a(x), \tag{3.6}$$

where the field $B^a_{\mu c}(x)$ is defined by

$$B^a_{\mu c}(x) \equiv -D^{\dagger ac}_{x,\mu}(U), \tag{3.7a}$$

$$B^a_{\mu c}(x) = G^{ac}_\mu(x) - G^{ac}_\mu(x-e_\mu) + 2^{-1}f^{abc}\left[A^b_\mu(x) + A^b_\mu(x-e_\mu)\right]. \tag{3.7b}$$

Here $A^b_\mu(x)$ and $G^{ac}_\mu(x)$ are linear combinations of the components of $U_\mu(x)$ and $U_\mu(x-e_\mu)$, given in eqs. (1.6) and (B.12a).

Remark. Recall that the restricted Gribov region $\Omega_{L,\mathrm{r}}$ is the set of configurations which are relative minima of the minimizing function, with vanishing zero-frequency component. This restriction eliminates the trivial null space of the Faddeev–Popov operator $M = -D(U)\cdot\nabla$ which consists of constant eigenvectors, so $M(U)$ is strictly positive and invertible in the interior of $\Omega_{L,\mathrm{r}}$. The quantity $B^a_{\mu c} = B^a_{\mu c}(U)$ is orthogonal to this null space for $U\in\Omega_{L,\mathrm{r}}$ because, with $A = A(U)$, we have

$$\sum_x B^a_{\mu c}(x) = \sum_x f^{abc}A^b_\mu(x) = 0.$$

The inverse $[M(U)^{-1}B]^a_{\mu c}(x)$ is taken in the orthogonal space, so $\omega^a(x)$ has a vanishing zero-frequency component,

$$\sum_x \omega^a_{(1)}(x) = 0. \tag{3.8}$$

Proof of lemma 1. We make use of the elementary relations derived in Appendix B, and the notation defined there, according to which, for the gauge transformation $h(x) = \exp[\varepsilon\omega_{(1)}(x)]$, the transformed quantity $(A^h) \equiv A(U^h)$ is given to first order in ε by

$$t^c(A^h)^c_\mu(x) = t^c A^c_\mu(x) + \varepsilon t^c\big[D_\mu(U)\omega_{(1)}\big]^c(x).$$

To this order, the transversality condition (2.10a) reads,

$$t^c\big[\nabla\cdot D(A)\omega_{(1)}\big]^c(x) + 2^{-1}\sum_\mu\big[U_\mu(x)t^c\eta^c_{(1)\mu} - t^c\eta^c_{(1)\mu}U_\mu(x-e_\mu)$$

$$-\,\mathrm{h.c.}\big]_{\text{traceless}} = 0,$$

where $\nabla\cdot$ represents the lattice divergence defined in eq. (B.6). For the SU(n) group, the anti-hermitian generators t^a form a complete basis for traceless matrices that satisfy $2\,\mathrm{tr}(t^a t^c) = -\delta^{ac}$, so we may rewrite the last equation in the form

$$-\big[\nabla\cdot D(A)\omega_{(1)}\big]^a(x) - B^a_{\mu c}(x)\eta^c_{(1)\mu} = 0, \tag{3.9}$$

where $B^a_{\mu c}(x)$ is given in eq. (3.7). This is seen from

$$2\,\mathrm{Re}\,\mathrm{tr}\big\{t^a\big[U_\mu(x)t^c - t^c U_\mu(x-e_\mu)\big]\big\}$$

$$= \mathrm{Re}\,\mathrm{tr}\big[(\{t^a, t^c\} - [t^a, t^c])U_\mu(x)$$

$$-(\{t^a, t^c\} + [t^a, t^c])U_\mu(x-e_\mu)\big]$$

$$= \mathrm{Re}\,\mathrm{tr}\big\{\{t^a, t^c\}\big[U_\mu(x) - U_\mu(x-e_\mu)\big]$$

$$+f^{abc}t^b\big[U_\mu(x) + U_\mu(x-e_\mu)\big]\big\},$$

which agrees with the expression for $-B^a_{\mu c}(x)$ given in eq. (3.7b). QED

To locate the distinguished boundary we must compare the minimizing function $J(U^h, \theta)$, defined in eq. (2.18), with $I(U)$ for small θ. This comparison is simplified by the following lemma.

Lemma 2. Let U be a configuration in $\Omega_{L,r}$, and let $\eta \equiv \eta_{(1)}$ and $\omega_{(1)}$ satisfy eqs. (3.3) and (3.4) respectively. Then to order ε^3, the minimizing function $J(U^h, \theta)$ is given by

$$J(U^h, \theta) - I(U) = -\varepsilon^2 (2n)^{-1} H(U, \eta^*) + \varepsilon^3 C(U, \eta^*) + \dots, \tag{3.10}$$

where $H(U, \eta)$ and $C(U, \eta)$ are respectively quadratic and cubic in η, $H(U, \eta)$ is given explicitly in eqs. (3.11) and (3.12) below, and η^* is chosen to maximize $H(U, \eta)$, eq. (3.13), below.

We call "the horizon tensor", the tensor $H_{\mu a, \nu b}(U)$ associated with the quadratic form $H(U, \eta)$,

$$H(U, \eta) = \eta_\mu^a H_{\mu a, \nu b}(U) \eta_\nu^b, \tag{3.11}$$

defined by

$$H_{\mu a, \nu b}(U) \equiv \sum_{x,y} D_{x,\mu}^{ac}(U) D_{y,\nu}^{bd}(U) M_{x,c;y,d}^{-1}(U)$$
$$- \delta_{\mu\nu} \sum_x G_\mu^{ab}(x), \tag{3.12a}$$

or

$$H_{\mu a, \nu b}(U) \equiv \left(B_{\mu a}, M^{-1} B_{\nu b}\right) - \delta_{\mu\nu} \sum_x G_\mu^{ab}(x), \tag{3.12b}$$

where the real fields $B_{\mu c}^a(x)$ and $G_\mu^{ab}(x)$ are given in eqs. (3.7) and (B.12). The Lorentz–color vector η^* maximizes $H(U, \eta)$,

$$H(U, \eta^*) \geqslant H(U, \eta) \quad \text{for all } \eta \text{ and } \eta^* \text{ satisfying (3.3)}. \tag{3.13}$$

Proof of lemma 2. In the expansion of $J(U^h, \theta)$, there is no term linear in $\omega_{(1)}$ and $\eta_{(1)}$ because U is a transverse configuration with vanishing zero-frequency component. For the same reason, the minimizing function $J(U^h, \theta)$ to order ε^2, is independent of the quantities $\eta_{(2)}$ and $\omega_{(2)}$ defined in eqs. (3.1) and (3.2). Expressions (3.11) and (3.12) for $H(U, \eta_{(1)})$ are found from eqs. (3.1) and (3.2) by straightforward expansion in powers of ε, with $\eta_{(2)} = \omega_{(2)} = 0$. Consequently, with η^* chosen to satisfy eq. (3.13), the minimizing function given by eqs. (3.10) to (3.13) is correct to order ε^2. QED

Remark. To order ε^3, the minimizing function $J(U^h, \theta)$ is independent of the quantities $\eta_{(3)}$ and $\omega_{(3)}$ because U is a transverse configuration with vanishing zero-frequency component. Moreover, to order ε^3, it is also independent of $\omega_{(2)}$

and $\eta_{(2)}$ because, with $g(x) = \exp[\varepsilon\omega(x)] \exp(t^a\theta^a_\mu)$, the configuration U^g is transverse to order ε by virtue of lemma 1, and with η^* satisfying eq. (3.13), the stationarity condition (2.21) is also satisfied to order ε. Therefore the quantity $C(U, \eta^*)$ may also be calculated by straightforward expansion in powers of ε, with $\eta_{(2)} = \omega_{(2)} = \eta_{(3)} = \omega_{(3)} = 0$. Its explicit form is not needed however.

Theorem 2. Let η and $H(U, \eta)$ and $C(U, \eta)$ and a maximizing Lorentz–color vector η^* be defined as in lemma 2 above.

(i) Let U be a configuration in the core region, $U \in \Xi_L$, then the inequality

$$H(U, \eta) \leqslant 0 \tag{3.14}$$

holds.

(ii) Let U_0 be a configuration on the distinguished boundary of Ξ_L. Then the quadratic term vanishes for some maximizing $\eta^* = \eta^*(U_0)$,

$$H(U_0, \eta^*) = 0, \tag{3.15}$$

and moreover the cubic term vanishes for all such η^*,

$$C(U_0, \eta^*) = 0. \tag{3.16}$$

Remark 1. Properties (ii) are necessary conditions for a configuration U to lie on the distinguished boundary of Ξ_L. The remaining sufficient conditions that must also hold for a configuration U to lie on the distinguished boundary of Ξ_L are merely qualitative in nature, such as, for example, that various fourth derivatives should be positive and sufficiently large.

Remark 2. As a special case on the distinguished boundary the quadratic term may vanish identically,

$$H_{\mu a, \nu b}(U_0) = 0. \tag{3.17}$$

In this case, according to theorem 2, the cubic term vanishes identically,

$$C(U_0, \eta) = 0, \tag{3.18}$$

for all η satisfying eq. (3.3). We call the subset of the distinguished boundary where the last two conditions hold "the extreme boundary". Whereas examples of configurations on the distinguished boundary are given in Appendices C and E, we have no examples of points on the extreme boundary at finite L. It is possible that the extreme boundary is empty at finite L. However, in the next section we shall be interested in a distribution that gets concentrated on the extreme boundary in the infinite-L limit.

Proof of theorem 2. Inequality (3.14) holds for any η because of the minimizing property, $J(U^h, \theta) \geqslant I(U)$ for all h and θ, when $U \in \Xi_L$, and because for sufficiently small ε, the difference $J(U^h, \theta) - I(U)$ has the sign of the quadratic piece, so statement (i) is established. To prove (ii), observe that if U_0 lies on the distinguished boundary, then along a curve $U(t)$ that crosses this boundary U_0, the difference $J(U^h, \theta) - I(U)$ changes sign for arbitrarily small ε when $U(t) = U_0$. With $J(U^h, \theta)$ given in eq. (3.10), this can happen only if the term which is quadratic in ε vanishes at $U = U_0$. This proves eq. (3.15). For U_0 to lie on the distinguished boundary, the absolute minimum of the minimizing gauge transformation must occur at $\varepsilon = 0$. However, if in eq. (3.10), the quadratic term $H(U_0, \eta^*)$ vanishes and the cubic term $C(U_0, \eta^*)$ is non-zero then $J(U_0^h, \theta) - I(U_0)$ is negative for some non-zero ε, in which case U_0 does not lie in Ξ_L, which is absurd. QED.

4. Horizon function

In this section we derive some properties of the core region summarized in theorem 3 below.

We have seen in theorem 2, that $H_{\mu a, \nu b}(U)$ is bounded from above. According to the following lemma, it is also bounded from below.

Lemma. (i) For all U in the restricted Gribov region $\Omega_{\mathrm{L,r}}$, the horizon tensor satisfies

$$-V v_\mu^a v_\mu^a \leqslant v_\mu^a H_{\mu a, \nu b}(U) v_\nu^b \tag{4.1}$$

for all Lorentz–color vectors v_μ^a.

(ii) For all U in $\Omega_{L,\mathrm{r}}$, the horizon tensor has the form

$$H_{\mu a, \nu b}(U) = -V \delta_{\mu\nu} \delta_{ab}, \tag{4.2}$$

if and only if U is the vacuum configuration, $U_\mu(x) = 1$.

(iii) For all U in the restricted Gribov region $\Omega_{L,\mathrm{r}}$, the horizon tensor $H_{\mu a, \nu b}(U)$ is regular except for points U on the boundary $\partial \Omega_{L,\mathrm{r}}$.

Proof of the lemma. The first term in eq. (3.12) is manifestly non-negative because, by definition, $\Omega_{L,\mathrm{r}}$ is the region where $M(U)$ is a positive matrix. Thus, to prove eq. (4.1) it is sufficient to prove that the color tensor $\delta_{ab} - G_\mu^{ab}(x)$ is non-negative for each x and μ. Consider the expression

$$Q \equiv -\tfrac{1}{2} v^a v^b \operatorname{tr}\left[(t^a t^b + t^b t^a)(1 - U^\dagger)(1 - U)\right],$$

where U is an SU(n) matrix, and the t^a are an anti-hermitian basis of the SU(n) Lie algebra with $\mathrm{tr}(t^a t^b) = -\frac{1}{2}\delta^{ab}$. With $U = U_\mu(x)$ we have

$$Q = v^a v^b \left[\delta^{cd} - G_\mu^{cd}(x)\right],$$

by eq. (B.12a), so it is sufficient to prove that Q is non-negative for all color vectors v^a. Let v represent the anti-hermitian matrix $v \equiv v^a t^a = -v^\dagger$. We have

$$Q = \mathrm{tr}\left[v^\dagger(1-U^\dagger)(1-U)v\right] \geqslant 0,$$

so (i) is established. Moreover, this last expression vanishes for all v only if $U = 1$, so $\delta^{cd} - G_\mu^{cd}(x)$ vanishes for all x and μ if and only if $U_\mu(x) = 1$ for all x and μ. Moreover, if $U_\mu(x) = 1$ for all x and μ, then, according to eq. (3.7b), $B_{\mu c}^a(x)$ vanishes, and with it the first term in eq. (3.12), so (ii) is established. Item (iii) is obvious. QED.

We call "the horizon function", the trace $H(U)$ of the horizon tensor,

$$H(U) \equiv \sum_{\mu,a} H_{\mu a,\mu a}(U), \tag{4.3}$$

$$H(U) = \sum_{x,y} D_{x,\mu}^{ac}(U) D_{y,\mu}^{ad}(U) M_{x,c;y,d}^{-1}(U)$$
$$-(n^2-1)E(U), \tag{4.4a}$$

where $E(U)$ is the maximizing function (1.2b), and the gauge-covariant derivative is given in eq. (B.11). Its expansion in eigenvectors $\omega_n^a(x)$ of the lattice Faddeev–Popov matrix $M(U)$ is given by

$$H(U) \equiv \sum_{n,\mu,a} \left[\lambda_n^{-1}\left(\sum_x \left[D_\mu(U)\omega_n\right]^a(x)\right)^2 - (n^2-1)E(U)\right]. \tag{4.4b}$$

We also define the horizon function per lattice site,

$$h(U) \equiv H(U)/V. \tag{4.5}$$

Theorem 3. (i) for all U in the core region Ξ_L, the horizon function per lattice site $h(U)$ lies in the interval

$$-f \leqslant h(U) \leqslant 0, \tag{4.6}$$

where

$$f \equiv (n^2-1)D \tag{4.7}$$

is the number of degrees of freedom per lattice site.

(ii) For all U in Ξ_L, the upper bound is saturated,

$$h(U) = 0, \tag{4.8}$$

if and only if U lies in the extreme boundary of Ξ_L.

(iii) For all U in the restricted Gribov region $\Omega_{L,\mathrm{r}}$, the lower bound is saturated,

$$h(U) = -f, \tag{4.9}$$

if and only if U is the vacuum configuration $U_\mu(x) = 1$.

(iv) For all in $\Omega_{L,\mathrm{r}}$, the value

$$h(U) = \infty \tag{4.10}$$

is achieved if and only if U lies on the boundary $\partial\Omega_{L,\mathrm{r}}$, apart from a lower dimensional set on $\partial\Omega_{L,\mathrm{r}}$, where the value of $h(U)$ depends on how the boundary point is approached.

Proof of theorem 3. Consider the color vector $\eta_\mu^a = n_\mu v^a$, where n_μ and v^a are both normalized to unity. It satisfies eq. (3.3). By choosing n_μ and v^a to both lie along principal axes in Lorentz and color space, we find from eqs. (4.1) and (3.14) that for U in Ξ_L, the diagonal elements of $H_{\mu a,\nu b}(U)$ satisfy

$$-V \leqslant H_{\mu a,\mu a}(U) \leqslant 0 \quad \text{(no summation)} \tag{4.11}$$

for each μ and a. Summation over μ and a gives eq. (4.6). This proves (i). To prove (ii), assume first that U lies on the extreme boundary, so eq. (3.17) holds. Upon taking the trace of eq. (3.17), we obtain $H(U) = 0$. Conversely assume that $U \in \Xi_L$ and $H(U) = 0$. Then, by eq. (4.11), it follows that the diagonal elements of $H_{\mu a,\nu b}(U)$ satisfy $H_{\mu a,\mu a}(U) = 0$ for each μ and a. In this case however the off-diagonal elements of $H_{\mu a,\nu b}(U)$ must vanish, for otherwise the upper bound (3.14) would be violated. Thus we have $H_{\mu a,\nu b}(U) = 0$, which holds only on the extreme boundary of Ξ_L. This establishes (ii). To prove (iii), assume first that U is the vacuum configuration $U = 1$. Then by the preceding lemma $H_{\mu a,\nu b}(U) = -V\delta_{\mu\nu}\delta_{ab}$, which implies $H(U) = -fV$. Conversely, assume $U \in \Omega_{L,\mathrm{r}}$ and $H(U) = -fV$. Then by eq. (4.11), the diagonal elements of the horizon tensor $H_{\mu a,\nu b}(U)$ satisfy $H_{\mu a,\mu a}(U) = -V$. In this case however the off-diagonal elements also vanish, for otherwise the lower bound (4.1) would be violated, and we obtain $H_{\mu a,\nu b}(U) = -V\delta_{\mu\nu}\delta_{ab}$. By the preceding lemma, this is true in the restricted Gribov region $\Omega_{L,\mathrm{r}}$ only for $U = 1$. To prove (iv) observe that the boundary $\partial\Omega_L$ satisfies the equation $\det[M(U)] = 0$. The first term in the horizon function $(B, M^{-1}B)$ is indeterminate, of the form $0/0$, only when the equations $\det[M(U)] = 0$ and $(B, \operatorname{cof} M(U)B) = 0$ both hold, where $\operatorname{cof} M$ is the cofactor matrix. Therefore the points on the boundary that are indeterminate do lie on a lower dimensional set, as asserted, unless the two surfaces coincide. To exclude this last possibility, it is sufficient that they be distinct in some neighborhood. In Appendices C and E, configurations U on $\partial\Omega_{L,\mathrm{r}}$ are exhibited, for which $M(U)$ is explicitly invertible. As it happens, these configurations do lie on both surfaces, and in fact, for the configuration exhibited in Appendix E, the stronger condition $B(U)=0$ holds. The reason is that these configurations were chosen because exact calculations are possible, and in fact it may be verified that the two surfaces are distinct in a neighborhood of these configurations, apart from a lower dimensional set. QED.

Remark 1. This theorem may be restated pictorially as follows. Consider the subset of the restricted Gribov region $\Omega_{L,\mathrm{r}}$ defined by the condition

$$H(U) \leqslant E. \tag{4.12}$$

As E increases from $-V$ to $+\infty$, this region grows, as if a diaphragm were opening, from the vacuum configuration alone at $E=-V$ to finally include all of $\Omega_{L,\mathrm{r}}$ at $E=\infty$. Moreover, the region defined by $H(U)\leqslant 0$ includes the core region Ξ_L, and among configurations U in Ξ_L, the value $H(U)=0$ is achieved, if at all, only on the extreme boundary of Ξ_L.

It is convenient to define the region just mentioned,

$$\Psi_L \equiv \{U\colon H(U)\leqslant 0\} \cap \Omega_{L,\mathrm{r}}, \tag{4.13}$$

which we call "the augmented core". The inclusions

$$\Xi_L \subset \Psi_L \subset \Omega_{L,\mathrm{r}} \tag{4.14}$$

are obvious. From item (ii) of theorem 3, we observe that the extreme boundary of the core is the intersection of the boundaries,

$$(\text{extreme boundary of } \Xi_L) = \partial\Xi_L \cap \partial\Psi_L. \tag{4.15}$$

Let U be a configuration for which the equation

$$\left[D_\mu^{ac}(U)\omega^c\right](x)=0 \tag{4.16}$$

possesses a non-zero solution ω of period L. An explicit example is exhibited in Appendix E. Such a configuration lies on a degenerate gauge orbit because, with $h(x)=\exp[\omega(x)]$, we have $U^h=U$. If eq. (4.15) possesses a non-zero solution, then for every local gauge transformation g_x, the equation $[D_\mu^{ac}(U^g)\omega'^c](x)=0$ also possesses a non-zero solution $\omega'(x)=g^\dagger(x)\omega(x)g(x)$, so this condition is gauge invariant. Any solution ω to eq. (4.16) is also a null eigenvector of the lattice Faddeev–Popov operator $M(U)$, namely, $[M^{ac}(U)\omega^c](x)=0$, because $M=-\nabla\cdot$

$D(U)$. Consequently a degenerate gauge orbit can only intersect Ω_L at a point U on its boundary $\partial\Omega_L$. This degenerate gauge orbit must also intersect the fundamental modular region Λ_L which is contained in Ω_L. Consequently the boundaries $\partial\Lambda_L$ and $\partial\Omega_L$ touch each other at the configuration U. On the other hand, for such a configuration, the first term of the horizon function is of the indeterminate form $0/0$, as one sees from eq. (4.4b). It follows that this same configuration U also lies on the lower dimensional set where the boundaries $\partial\Psi_L$ and $\partial\Omega_L$ also touch each other,

$$U \in \partial\Lambda_L \cap \partial\Psi_L \cap \partial\Omega_L. \tag{4.17}$$

We conclude with the observation that the fundamental modular region Λ_L and the augmented core Ψ_L are qualitatively similar at *finite* L. Both are contained in the Gribov region Ω_L, and both touch its boundary $\partial\Omega_L$ in a lower dimensional set which has at least some points in common, namely all points that lie on degenerate gauge orbits. This is quite remarkable, because the core Ξ_L, from which the augmented core Ψ_L is obtained, is only expected to approach Λ_L at *infinite* L. Moreover, the bound on all U in Ξ_L, namely $H(U) \leqslant 0$, which is the condition used to define the augmented core Ψ_L, comes from considering only (large) gauge transformations that lie at the zero-frequency limit of the Brillouin zone. Apparently the finite-frequency gauge transformations do not make a qualitative difference in the shape of the boundary even at finite L.

5. Saddle point and Boltzmann factor

Recall that the partition functions $Z(\Lambda_L)$ and $Z(\Xi_L)$ are obtained by integrating with the Faddeev–Popov weight over Λ_L and Ξ_L. Similarly, we define the partition function $Z(\Psi_L)$ to be the integral with the lattice Faddeev–Popov weight over the augmented core region Ψ_L,

$$Z(\Psi_L) \equiv \int_\Omega \mathrm{d}U\, \theta[-H(U)]\, \exp[-\beta S_{\mathrm{W}}(U)]\delta(\nabla\cdot A)\, \det[M(U)], \tag{5.1}$$

where the restriction to the Gribov region $\Omega = \Omega_L$ is necessary because there are some configurations outside Ω_L which also satisfy $H(U) \leqslant 0$. This formula defines the measure $\mathrm{d}\mu$,

$$Z(\Psi_L) = \int \mathrm{d}\mu\, \theta[-H(U)]. \tag{5.2}$$

In this section we shall show that the cutoff at the boundary $\partial\Psi_L$, represented by $\theta[-H(U)]$ may be replaced by a Boltzmann factor $\exp[-\alpha H(U)]$ in the infinite-volume limit, in the sense that the corresponding partition functions satisfy

$$\lim_{V\to\infty} Z(\Psi_L) = \lim_{V\to\infty} Z(\alpha), \tag{5.3}$$

provided that the thermodynamic parameter α has a particular value α_{cr}, obtained below. In sect. 7, we shall show that $Z(\alpha_{cr})$ and $Z(\Xi)$ are equal in the thermodynamic limit, if hypothesis (ii), which is explained there, is granted.

We represent the step-function $\theta[-H(U)]$ by its Fourier decomposition

$$Z(\Psi_L) = (2\pi)^{-1} \int \mathrm{d}\mu \int_{-\infty}^{\infty} \frac{\mathrm{d}y}{\varepsilon + iy} \exp[-iyH(U)]. \tag{5.4}$$

The integrand may be analytically continued as function of the complex variable $z \equiv x + iy$. We freely deform the contour of integration from $x = 0$ into the right-hand half-plane $x \geqslant 0$, where the integrand is analytic, and we write

$$Z(\Psi_L) = (2\pi i)^{-1} \int \frac{\mathrm{d}z}{z} \exp[W(z)] = (2\pi i)^{-1} \int \mathrm{d}z \, \exp[\Phi(z)], \tag{5.5}$$

where $W(z)$ is defined by

$$\exp[W(z)] \equiv \int \mathrm{d}\mu \, \exp[-zH(U)], \tag{5.6}$$

and

$$\Phi(z) \equiv W(z) - \ln z. \tag{5.7}$$

The exponent $zH(U)$ is of order V. We look for a saddle point which will dominate the integral in the infinite-volume limit. If there is a saddle point, it satisfies

$$\Phi'(z_{cr}) = W'(z_{cr}) - z_{cr}^{-1} = 0. \tag{5.8}$$

Theorem 4. The last equation has a unique positive real solution α_{cr}.

Remark 1. It will be found that α_{cr} is a thermodynamic parameter that has the dimension of $(\text{mass})^4$ in the continuum limit.

Proof of theorem 4. The condition for a stationary point reads

$$\Phi'(\alpha) = W'(\alpha) - \alpha^{-1} = -\langle H(U) \rangle - \alpha^{-1} = 0. \tag{5.9}$$

We also have

$$\begin{aligned} \Phi''(\alpha) &= W''(\alpha) + \alpha^{-2} \\ &= \langle H^2(U) \rangle - \langle H(U) \rangle^2 + \alpha^{-2} > 0, \end{aligned} \tag{5.10}$$

so $\Phi'(\alpha)$ is monotonically increasing. For U in Ω_L, the horizon function $H(U)$ is bounded below, so by eq. (5.9) we have $\Phi'(0) = -\infty$. The partition function $Z(\alpha)$ for real α is defined by means of the Boltzmann distribution,

$$Z(\alpha) = \exp[W(\alpha)] \equiv \int d\mu \, \exp[-\alpha H(U)]. \tag{5.11}$$

At $\alpha = +\infty$, the measure is dominated by the lower bound of $H(U)$ so we have $\langle H(U) \rangle = -V$ in this case, which gives $\Phi'(+\infty) = V$. Thus the equation $\Phi'(\alpha) = 0$ indeed has one real positive solution, as asserted. QED.

Remark 2. We have $W'(z) = -\langle H(U) \rangle_z$. If z has a non-zero imaginary part, $\langle H(U) \rangle_z$ is strongly suppressed due to rapid oscillations of $\exp(-zH)$, because $H(U)$ is of order V. Thus at large V there is no solution to the saddle point equation (5.8) for z_{cr} with non-zero imaginary part. Consequently the unique real saddle point, whose existence is asserted in theorem 4, is the only one.

In the representation for $Z(\Psi_L)$ given by the contour integral (5.5), the contour does cross the positive x axis. We choose a contour so this crossing occurs where eq. (5.8) is satisfied, namely where α has the unique value α_{cr} determined by

$$\langle H(U) \rangle = -\alpha^{-1}. \tag{5.12}$$

The sign of $\Phi''(\alpha) > 0$ assures that the contour crosses in the direction of steepest descent, and the contribution from this saddle point is given by

$$\begin{aligned} Z_s(\Psi_L) &= (2\pi)^{-1} \int_{-\infty}^{\infty} dv \, \exp[\Phi(\alpha_{cr} + iy)]_s, \\ &= (2\pi)^{-1} \exp[W(\alpha_{cr}) - \ln \alpha_{cr}] \int_{-\infty}^{\infty} dy \, \exp[-2^{-1}\Phi''(\alpha_{cr}) y^2] \\ &= [2\pi\Phi''(\alpha_{cr})]^{-1/2} \alpha_{cr}^{-1} \exp[W(\alpha_{cr})]. \end{aligned}$$

By eq. (5.10), we have $\Phi''(\alpha_{cr}) = O(V)$. Thus the width of the saddle vanishes like $V^{-1/2}$ in the infinite-volume limit, and we conclude that the contribution from the saddle point becomes exact in the infinite-volume limit,

$$\lim_{V \to \infty} Z(\Psi_L) = \lim_{V \to \infty} Z(\alpha_{cr}). \tag{5.13}$$

We have shown that the partition functions $Z(\Psi)$ and $Z(\alpha_{cr})$ are equal in the infinite-volume limit. This also implies the equality of all expectation values of local observables in the two distributions, because they may be calculated, as usual, by adding arbitrarily small local sources to the action. The saddle point, which is determined by an exponent that is of order V, gets shifted by the local sources by an amount which is negligible in the infinite-volume limit.

6. Local action and local horizon condition

The partition function $Z(\alpha)$, defined in eq. (5.11), is expressed as an integral over non-local quantities including, in particular, the Boltzmann factor $\exp[-\alpha H(U)]$, where $H(U)$ is non-local because of the appearance of M^{-1} in $H(U)$, eq. (1.13). The Boltzmann factor may be made local by representing it as an integral over auxiliary fields.

We use the identity

$$1 = \int \mathrm{d}\varphi\ \delta[M(U)\varphi - B]\ \det{}^f[M(U)], \tag{6.1}$$

where φ is a field that obeys the same transformation laws and has the same number of components $\varphi^a_{\mu c}(x)$ as does $B^a_{\mu c}(x)$. Thus $\varphi^a_{\mu c}(x)$ is a site variable, with preferred direction μ. The integration $\mathrm{d}\varphi$ and the delta-function refer to each component and position of the field $\varphi^a_{\mu c}(x)$. The delta-function imposes the constraint

$$[M(U)\varphi]^a_{\mu c}(x) = -\nabla \cdot [D(U)\varphi]^a_{\mu c}(x) = B^a_{\mu c}(x), \tag{6.2}$$

where $D_\lambda(U)$ is the lattice gauge-covariant derivative, eq. (B.11), and

$$\begin{aligned}[D_\lambda(U)\varphi]^a_{\mu d}(x) &= G^{ac}_\lambda(x)\left[\varphi^c_{\mu d}(x+e_\mu) - \varphi^c_{\mu d}(x)\right] \\ &\quad + 2^{-1} f^{abc} A^b_\lambda(c)\left[\varphi^c_{\mu d}(x+e_\mu) + \varphi^c_{\mu d}(x)\right].\end{aligned} \tag{6.3}$$

The indices μ and c are mute in eq. (6.2). They take on a total of $f = (n^2-1)D$ values, which explains the power f of the determinant in eq. (6.1). The delta-function may be expressed in terms of its Fourier decomposition,

$$1 = \int \mathrm{d}\varphi\ \mathrm{d}\bar{\varphi}\ \exp[(\bar{\varphi}, M(U)\varphi) - (\bar{\varphi}, B)]\ \det{}^f[M(U)].$$

Here $\bar{\varphi}$ is a field that has the same transformation laws and the same number of components, $\bar{\varphi}^a_{\mu c}(x)$, as $\varphi^a_{\mu c}(x)$. The integration over $\bar{\varphi}^a_{\mu c}(x)$ extends along the imaginary axis, so $\bar{\varphi}^a_{\mu c}(x)$ is *not* the complex conjugate of $\varphi^a_{\mu c}(x)$, but rather an independent Lagrange-multiplier field that enforces the constraint $M(U)\varphi = B$. The factor $\det{}^f[M(U)]$ may be expressed as an integral over pairs of independent

Grassmann variables $\omega^a_{\mu c}(x)$ and $\bar{\omega}^a_{\mu c}(x)$, that have the same number of components as $\varphi^a_{\mu c}(x)$ and $\bar{\varphi}^a_{\mu c}(x)$.

We thus obtain the local expression for the Boltzmann factor,

$$\exp[-\alpha H(U)] = \exp\{-\alpha[(B, M^{-1}B) - (n^2 - 1)E(U)]\},$$

$$\exp[-\alpha H(U)] = \exp[\alpha(n^2 - 1)E(U)] \int \mathrm{d}\varphi\, \mathrm{d}\bar{\varphi}\, \mathrm{d}\omega\, \mathrm{d}\bar{\omega}$$
$$\times \exp[(\bar{\varphi}, M(U)\varphi) - (\bar{\omega}, M(U)\omega) - \alpha(B, \varphi) - (\bar{\varphi}, B)]. \tag{6.4a}$$

A change of variables by the rescaling $\varphi' = \alpha^{-1/2}\varphi$ and $\bar{\varphi}' = \alpha^{1/2}\bar{\varphi}$ gives the alternative form

$$\exp[-\alpha H(U)] = \exp[\alpha(n^2 - 1)E(U)] \int \mathrm{d}\varphi\, \mathrm{d}\bar{\varphi}\, \mathrm{d}\omega\, \mathrm{d}\bar{\omega}$$
$$\times \exp[(\bar{\varphi}, M(U)\varphi) - (\bar{\omega}, M(U)\omega) - \alpha^{1/2}(B, \varphi + \bar{\varphi})], \tag{6.4b}$$

in which the exponent is symmetric under the interchange $\varphi \leftrightarrow \bar{\varphi}$. Although this symmetry is violated by the path of integration, which is real for φ and pure imaginary for $\bar{\varphi}$, nevertheless, the easily verified formula

$$\int \mathrm{d}\varphi\, \mathrm{d}\bar{\varphi}\, \exp[(\bar{\varphi}, M\varphi) + (K, \varphi) + (\bar{\varphi}, J)] = \exp[-(K, M^{-1}J)], \tag{6.5}$$

where J and K are independent real sources, shows that the formal hermiticity of the action under the interchange $\varphi \leftrightarrow \bar{\varphi}$ is in fact realized in expectation values. This formula also shows that the φ and $\bar{\phi}$ fields have a negative propagator $-M^{-1}$ and thus are pure (bose) ghosts. Feynman rules for the fields φ, $\bar{\varphi}$, ω and $\bar{\omega}$ may be derived in the standard way from Wick's theorem.

By eq. (3.7), the last term in eq. (6.4) may be rewritten in terms of the gauge-covariant derivative,

$$(B, \varphi + \bar{\varphi}) = -(D(U)(\varphi + \bar{\varphi}))$$
$$\equiv -\sum_{x,\mu,c} [D_\mu(U)(\phi + \bar{\phi})]^c_{\mu c}(x). \tag{6.6}$$

The lattice Faddeev–Popov weight $\mathrm{d}\mu$, defined by eqs. (5.1) and (5.2), may be represented by an integration over the familiar Faddeev–Popov auxiliary fields,

$$\int \mathrm{d}\mu \equiv \int_\Omega \mathrm{d}U\, \mathrm{d}\lambda\, \mathrm{d}C\, \mathrm{d}C^*$$
$$\times \exp[-\beta S_{\mathrm{W}}(U) - (\lambda, \nabla\cdot A) - (C^*, M(U)C)]. \tag{6.7}$$

Here $C^a(x)$ and $C^{*a}(x)$ are the standard pair of lattice Faddeev–Popov Fermi ghosts, and $\lambda^a(x)$ is a Lagrange-multiplier field that is integrated along the imaginary axis, and which enforces the constraint $\nabla \cdot A^a(x) = 0$. Before integration over $\lambda^a(x)$, the field $A^a_\mu(x)$ is not transverse which means that $M(U)$, as defined in eq. (B.18), is not symmetric. We stipulate that $M(U)C$ is given by

$$[M(U)C]^a(x) \equiv -\nabla \cdot [D(U)C]^a(x). \tag{6.8}$$

This gives for the partition function

$$Z(\alpha) = \int_\Omega \mathrm{d}U\, \mathrm{d}\lambda\, \mathrm{d}C\, \mathrm{d}C^*\, \mathrm{d}\varphi\, \mathrm{d}\bar{\varphi}\, \mathrm{d}\omega\, \mathrm{d}\bar{\omega}\, \exp(-S_{\mathrm{loc}}), \tag{6.9}$$

where

$$\begin{aligned} S_{\mathrm{loc}} \equiv \beta S_{\mathrm{W}}(U) &+ (\lambda, \nabla \cdot A) + (C^*, M(U)C) - (\bar{\varphi}, M(U)\varphi) \\ &+ (\bar{\omega}, M(U)\omega) - (D(U)\bar{\varphi}) - \alpha\big[(D(U)\varphi) + (n^2-1)E(U)\big], \end{aligned} \tag{6.10}$$

where $(D(U)\bar{\varphi})$ and $\alpha(D(U)\varphi)$ are defined in eq. (6.6). This action is local. Apart from the Wilson action $S_{\mathrm{W}}(U)$, it involves only the lattice gauge-covariant derivative $D(U)$ of the various ghost fields and the lattice divergence $\nabla \cdot$. We defer until sect. 8 a discussion of the physical significance of the various terms which appear in S_{loc}, in order to first establish the equality of $Z(\alpha_{\mathrm{cr}})$ and $Z(\Xi)$.

The locality of formula (6.9) for the partition function $Z(\alpha)$ is marred only by the restriction to the Gribov region $\Omega = \Omega_L$. However, the integrand vanishes at the boundary $\partial\Omega_L$ because of the Boltzmann factor $\exp[-\alpha H(U)]$ or the factor $\det[M(U)]$ in the Faddeev–Popov weight. Consequently the cutoff at $\partial\Omega_L$ does not appear at all in a differential characterization of the partition function. This will allow us in sect. 9 to obtain a purely local characterization of the lattice partition function $Z(\alpha)$ and of the lattice effective action Γ, by means of the Schwinger–Dyson equations, which depend only on the local action S_{loc}. Formulas (6.9) and (6.10) may also be used for a perturbative expansion, described in sect. 9 and Appendix H. The perturbation expansion is also local, because it is an expansion about $U_\mu(x) = 1$, so it is insensitive to the cutoff at $\partial\Omega_L$. However, because of the Boltzmann factor $\exp[-\alpha H(U)]$, this expansion reflects the cutoff at the boundary $\partial\Psi_L$ of the augmented core Ψ_L, which is contained in the Gribov region, $\Psi_L \subset \Omega_L$.

The equation $W'(\alpha) = \alpha^{-1}$, which determines the critical value α_{cr}, may also be expressed in terms of local fields. From eq. (6.10) this is equivalent to

$$\langle H_{\mathrm{loc}}(U) \rangle = -\alpha^{-1}, \tag{6.11}$$

where $H_{\mathrm{loc}}(U)$ now has the local expression

$$H_{\mathrm{loc}}(U) \equiv -(n^2-1)E(U) - (D(U)\varphi). \tag{6.12}$$

Translation invariance allows us to replace the left hand side by the expectation value of a local field. For this purpose we write

$$H_{\mathrm{loc}}(U) = \sum_x h(x), \tag{6.13}$$

where

$$h(x) \equiv -(n^2-1)\sum_\mu n^{-1} \operatorname{Re} \operatorname{tr}\left[U_\mu(x)\right] + \left[D_\mu(U)\varphi\right]^a_{\mu a}(x), \tag{6.14}$$

and repeated indices are summed over. The equation $W'(\alpha) = \alpha^{-1}$ is equivalent to the local equation

$$\langle h(x) \rangle = -(\alpha V)^{-1}. \tag{6.15}$$

7. Equality of $Z(\alpha_{\mathrm{cr}}) = Z(\Psi)$ and $Z(\Xi)$

We proved in sect. 5 that the equation $\langle h(x)\rangle = -(\alpha V)^{-1}$ determines a unique positive value of α. However, the left hand side of this equation remains finite in the infinite-volume limit for finite α. Thus $(\alpha V)^{-1}$ is negligible in the infinite-volume limit, and the equation which determines α_{cr} in the infinite-volume limit reads simply

$$\langle h(x) \rangle = 0. \tag{7.1}$$

We call this equation "the horizon condition" because $H(U) = 0$ is satisfied on the boundary of the augmented core region Ψ.

Recall that Ψ_L was defined by the condition $H \leqslant 0$, so the probability distribution $p(h)$ for the random variable $h(U) = H(U)/V$, vanishes for $h > 0$,

$$p(h) = 0 \quad \text{for } h > 0, \tag{7.2}$$

The last two equations are consistent only if $p(h)$ satisfies

$$\lim_{V\to\infty} p(h) = \delta(h). \tag{7.3}$$

Thus, in the infinite-volume limit, the probability distribution associated with the partition function $Z(\Psi) = Z(\alpha_{\mathrm{cr}})$ gets entirely concentrated on the boundary $\partial\Psi$ of the augmented core region where the condition $h(U) = 0$ is satisfied.

Eq. (7.3) is true only if the variance of $h(U)$ vanishes in the infinite-volume limit. To verify this point, we calculate the variance of $h(U)$ explicitly,

$$V^{-2}\big(\langle H^2\rangle - \langle H\rangle^2\big) = V^{-2}\sum_{x,y}\big[\langle h(x)h(y)\rangle - \langle h(0)\rangle^2\big]$$

$$= V^{-1}\sum_{x}\big[\langle h(x)h(0)\rangle - \langle h(x)\rangle\langle h(0)\rangle\big],$$

where the local field $h(x)$ is given in eq. (6.14), and we have again used translation invariance. This gives

$$V^{-2}\big(\langle H^2\rangle - \langle H\rangle^2\big) = V^{-1}\sum_{x} c(x), \tag{7.4}$$

where $c(x)$ is the connected correlation function of the field $h(x)$ in the distribution $Z(\alpha)$. This quantity does vanish in the infinite-volume limit provided only that $c(x)$ falls off at large distance. We may use the same method to prove a sharper result.

Theorem 5. In the infinite-volume limit, the probability distribution $Z(\Psi) = Z(\alpha_{\mathrm{cr}})$ associated with the augmented core region Ψ, gets concentrated on the submanifold of $\partial\Psi$ where the equations $h_{\mu a,\nu b}(U) = 0$ and $c_{\lambda a,\mu b,\nu c}(U) = 0$ hold. Here $h_{\mu a,\nu b}(U) \equiv H_{\mu a,\nu b}(U)/V$ and $c_{\lambda a,\mu b,\nu c}(U) \equiv C_{\lambda a,\mu b,\nu c}(U)/V$ have a finite limit for $V \to \infty$.

Proof of theorem 5. To prove the theorem, we first show that, in the infinite-volume limit, the stated conditions hold in the mean, $\langle H_{\mu a,\nu b}\rangle/V = \langle C_{\lambda a,\mu b,\nu c}\rangle/V = 0$, and then we show that the variances of these quantities also vanish in the infinite-volume limit. The method is to express $H_{\mu a,\nu b}$ and $C_{\lambda a,\mu b,\nu c}$ as sums over the volume of local fields.

Lattice and global color symmetries give

$$\big\langle H_{\mu a,\nu b}(U)\big\rangle = \mathrm{const.}\times\delta_{\mu\nu}\delta_{ab}. \tag{7.5}$$

The constant is fixed by the horizon condition, eq. (6.11), and we have

$$\langle H(U)_{\mu a,\nu b}\rangle/V = -\delta_{\mu\nu}\delta_{ab}(\alpha f V)^{-1}. \tag{7.6}$$

In the infinite-volume limit, the last term is negligible, as was noted previously, so the desired constraint, $H(U)_{\mu a,\nu b}/V = 0$, is satisfied in the mean.

We now verify that the variance of $H(U)_{\mu a,\nu b}/V$ vanishes in the infinite-volume limit. By eqs. (3.12) and (6.2), the tensor $H_{\mu a,\nu b}$ may be written

$$H_{\mu a,\nu b} = \sum_{x} h_{\mu a,\nu b}(x), \tag{7.7}$$

where $h_{\mu a,\nu b}(x)$ is the local field

$$h_{\mu a,\nu b}(x) \equiv -\left[D_\mu^{ac}(U)\varphi_{\nu b}^c\right](x) - \delta_{\mu\nu}G_\mu^{ab}(x). \tag{7.8}$$

To simplify the notation we temporarily write H_i and $h_i(x)$, where the single index i represents the above four indices. The calculation leading to eq. (7.4) gives

$$V^{-2}\left(\langle H_i^2\rangle + \langle H_i\rangle^2\right) = V^{-1}\sum_x c_i(x), \tag{7.9}$$

where $c_i(x)$ is the connected correlation function of the local field $h_i(x)$ in the distribution $Z(\alpha)$. As before, the quantity $V^{-1}\Sigma_x c_i(x)$ vanishes in the infinite-volume limit. Thus in the infinite-volume limit, the probability distribution $Z(\alpha_{\text{cr}})$ gets concentrated on the submanifold where $h_{\mu a,\nu b}(U) = 0$.

The conditions on $c_{\lambda a,\mu b,\nu c}(U)$ are treated similarly. By lattice symmetries, we have

$$\langle C(U)_i\rangle / V = 0. \tag{7.10}$$

where we have written $C(U)_i$ for $C(U)_{\lambda a,\mu b,\nu c}$. Next note that because the minimizing function is the sum over the volume of a density, it follows by eq. (3.10) that $C_i(U)$ may be expressed as a sum over the volume of a density $k_i(x)$,

$$C_i(U) = \sum_x k_i(x). \tag{7.11}$$

By eqs. (3.4) and (6.2), we see that, inside an expectation value, the quantity $\omega(x) \sim M^{-1}B(x)$ may be replaced by the local field $\varphi(x)$. When this is done, the density $k_i(x)$ is expressed in terms of local fields $U(x)$ and $\varphi(x)$. It then follows as for $H_i(U)/V$ that the variance of $C_i(U)/V$ vanishes in the infinite-volume limit. Thus in the distribution $Z(\alpha_{\text{cr}})$, the probability distribution gets concentrated on the submanifold where $C_i(U)/V \to 0$. QED.

Remark 1. Only the condition $h(U) \leqslant 0$ is imposed point-wise on configurations U in Ψ_L, where $h(U)$ is the trace of the horizon tensor $h_{\mu a,\nu b}(U)$. However, the more detailed point-wise (in U-space) conditions $h_{\mu a,\nu b}(U) = 0$ and $c_{\lambda a,\mu b,\nu c}(U) = 0$ are regained in the infinite-volume limit because of the statistics of large numbers.

Remark 2. Remarkably, these equations are precisely the conditions which hold on the intersection $\partial\Xi_L \cap \partial\Psi_L$. Recall that the inclusion $\Xi_L \subset \Psi_L$ is proper. Nevertheless, as noted in eq. (4.15), the intersection of the boundaries $\partial\Psi_L$ and $\partial\Xi_L$ constitutes the extreme boundary of Ξ_L, which is that part of the boundary $\partial\Xi_L$ where $H_{\mu a,\nu b}(U) = 0$ and $C_{\lambda a,\mu b,\nu c}(U) = 0$. (These tensors are the second and third derivatives of the minimizing function $J(U^h, \theta)$.)

As noted in sect. 3, remark 1 following theorem 2, there are other qualitative conditions which must also be satisfied for a configuration on the submanifold defined by these equations, to actually be a part of the extreme boundary of Ξ_L. For example the fourth derivatives of the minimizing function at zero frequency must be positive, for otherwise the minimizing function lies lower for part of the Brillouin zone at finite frequency. We are not in a position to verify whether these additional qualitative conditions are satisfied. Instead, we take as a working hypothesis that when the equations $H_{\mu a,\nu b}(U)=0$ and $C_{\lambda a,\mu b,\nu c}(U)=0$ are satisfied for some configuration U in the augmented core Ψ, then the additional qualitative conditions for a configuration to lie on the extreme boundary of Ξ are also satisfied. (This is the hypothesis (ii) mentioned in the introduction.) With this hypothesis, then, in the infinite-volume limit, $Z(\Psi)$ and $Z(\Xi)$ have a common support, namely the intersection $\partial\Xi \cap \partial\Psi$, which is the extreme boundary of Ξ. It follows that $Z(\Psi)$ vanishes on the difference set $\Delta \equiv \Psi - \Xi$, and we have

$$Z(\Psi)=\int_{\Psi}\mathrm{d}\mu=\int_{\Xi}\mathrm{d}\mu+\int_{\Delta}\mathrm{d}\mu=Z(\Xi)+\int_{\Delta}\mathrm{d}\mu=Z(\Xi). \tag{7.12}$$

From the previous results at infinite volume, $Z(\alpha_{\mathrm{cr}})=Z(\Psi)$ we have, in the infinite-volume limit, the desired equalities

$$Z(\alpha_{\mathrm{cr}})=Z(\Psi)=Z(\Xi)=Z(\Lambda). \tag{7.13}$$

The reader who is interested in applications may proceed directly to sect. 10.

8. BRS invariance

It has been shown recently that the naive continuum limit of the local lattice action S_{loc}, eq. (6.10), gives a continuum action which is perturbatively renormalizable, when defined, for example, by dimensional regularization, and that there are precisely two renormalization constants, just as in Faddeev–Popov perturbation theory in the Landau gauge [8]. This was done by exhibiting a BRS invariance of the continuum action obtained by setting α, the new thermodynamic parameter that comes from the Boltzmann factor, to zero. The α-dependent, BRS-violating terms are of dimension zero, because α has dimension $(\mathrm{mass})^4$, and are controlled by the method of external sources, as described in ref. [8]. In this section we shall show that the lattice action S_{loc} possesses the same property, namely that it possesses a BRS invariance when α is set to zero. This means that the continuum theory inherits this BRS symmetry from the lattice theory, and that renormalizability is a property of the lattice-regularized theory. The BRS invariance of the lattice Faddeev–Popov theory was exhibited by Neuberger [9]. This BRS invariance implies the familiar Slavnov identities for the effective action Γ that will be introduced in the next section, where the remaining non-locality caused by the restriction to the Gribov region Ω in formula (6.9) for the partition function $Z(\alpha)$ will also be eliminated.

When α is set to zero, the action is made up of the Wilson action $\beta S_W(U)$ plus a piece S_1 which comes from writing an expression of the form $\delta(F)\det(\partial F/\partial y)$ as an integral over local fields. It is a peculiarity of such a piece to be an exact form, $S_1 = s\psi$, where s is a nilpotent BRS operator, $s^2 = 0$. To exhibit this invariance, we saturate the Faddeev–Popov ghost field with Lie algebra generators,

$$C(x) \equiv C^a(x)t^a, \tag{8.1}$$

and define the BRS transformation s acting on lattice variables by

$$\begin{aligned} sU_\mu(x) &= -C(x)U_\mu(x) + U_\mu(x)C(x+e_\mu),\\ sC(x) &= -C^2(x),\\ sC^*(x) &= \lambda(x), \quad s\lambda(x) = 0,\\ s\varphi(x) &= \omega(x), \quad s\omega(x) = 0,\\ s\bar{\omega}(x) &= \bar{\varphi}(x), \quad s\bar{\varphi}(x) = 0. \end{aligned} \tag{8.2}$$

It is easily verified that s is nilpotent, $s^2 = 0$. The BRS transform of the link variable $U_\mu(x)$ has the form of an infinitesimal lattice gauge transformation. Consequently the Wilson action S_W is invariant under s,

$$sS_W(U) = 0. \tag{8.3}$$

For the same reason, the BRS transform $A_\mu^b(x)$ is given by $sA_\mu^b(x) = [D_\mu(U)C]^b(x)$, where $D_\mu(U)$ is the by-now familiar lattice gauge-covariant derivative.

Let the BRS-invariant action S_{inv} be defined by

$$S_{\mathrm{inv}} \equiv \beta S_W(U) + s\psi, \tag{8.4a}$$

where

$$\psi = \left[(C^*, \nabla\cdot A) - (\bar{\omega}, M(U)\varphi) - (D(U)\bar{\omega})\right]. \tag{8.4b}$$

We have

$$sS_{\mathrm{inv}} = 0, \tag{8.5}$$

and S_{inv} has the standard form of the gauge-invariant action $\beta S_W(U)$ plus a term which is an exact form $s\psi$. Keeping in mind that s anti-commutes with Grassmann fields, one finds

$$\begin{aligned} S_{\mathrm{inv}} = \beta S_W(U) &+ (\lambda, \nabla\cdot A) + (C^*, M(U)C) - (\bar{\varphi}, M(U)\varphi)\\ &+ (\bar{\omega}, M(U)\omega) - (D(U)\bar{\varphi}) - (D(sU)\bar{\omega}) + (\bar{\omega}, M(sU)\varphi), \end{aligned} \tag{8.6}$$

Because sU is linear in C, the last two terms are linear in C and have the form

$$-(D(sU)\bar{\omega}) + (\bar{\omega}, M(sU)\varphi) = (Y, C), \tag{8.7}$$

where the explicit expression for Y is not needed. Upon comparison with S_{loc}, we see from eqs. (6.9) and (6.10) that a translation of the integration variable C^* by $M^{-1}Y$, keeping other variables constant, brings S_{loc} to the form

$$S'_{\text{loc}} = S_{\text{inv}} - \alpha\big[(D(U)\varphi) + (n^2 - 1)E(U)\big]. \tag{8.8}$$

This change of variable is well defined, because, for a finite lattice, the integral that represents $Z(\alpha)$ is supported where $M(U)$ is a positive matrix. In terms of the new, translated variables, both the action and the BRS transformation are local, whereas in terms of the original untranslated variables, the BRS transformation acts non-locally.

Only the last term in S'_{loc} which is proportional to α, is not BRS invariant. Fortunately, it turns out that in the continuum limit the BRS-violating term has dimension zero. In ref. [8], renormalizability of the theory with arbitrary α was established by introducing local sources for the continuum fields corresponding to n^{-1} Re $\text{tr}[U_\mu(x)]$ and $[D_\lambda(U)\varphi]^a_{\mu c}(x)$, and for their BRS transforms, as well as for other elementary and composite fields.

We make some comments now on the physical significance of the terms in the action

$$S'_{\text{loc}} = S_{\text{inv}} + \alpha H_{\text{loc}} = \beta S_{\text{W}} + s\psi + \alpha H_{\text{loc}}. \tag{8.9}$$

(1) The presence of a BRS-violating term does not mean that some horrible gauge-violating error has been made. For gauge invariance was exploited and lost when the gauge was fixed whereas BRS invariance is a peculiarity of an action which comes from writing an expression of the form $\delta(F)\det(\partial F/\partial x)$ as an integral over local fields. Moreover, the theory with α as an arbitrary parameter, whose renormalizability is required, does not in fact represent a gauge theory at all, until α is assigned the critical value α_{cr}, determined by the horizon condition, but this value is known only after calculations in the more general theory with α arbitrary.

(2) Expectation values are of course well defined in lattice gauge theory with Wilson action βS_{W} and no gauge-fixing. The gauge is fixed as a tool to study the critical or weak-coupling limit, because without gauge-fixing there is no weak-coupling expansion and no renormalizable weak-coupling limit. A "gauge-fixing term" which gets added to the action must have the property that it does not change the expectation value of any gauge-invariant observable. It would of course be valuable to verify directly, if this were possible, that the term $s\psi - \alpha H$ has this gauge-fixing property. However, according to our derivation, this is true only when the parame-

ter α has the critical value α_{cr} that is determined by the horizon condition, and for this reason we cannot use familiar methods to verify it. According to the arguments of standard perturbative Faddeev–Popov theory, a term of the form $s\psi$ by itself does not change the expectation value of gauge-invariant objects. However, at the non-perturbative level it appears that if this term alone (without the BRS-violating term αH_{loc}) were added to the Wilson action, then all Gribov copies including the one in the fundamental modular region, having opposite signs, would cancel each other out completely, so that the lattice partition function would vanish identically [10]. In the absence of a verification that the gauge-fixing term $s\psi + \alpha H_{loc}$ leaves expectation values of all gauge-invariant observables unchanged, we content ourselves with the following observation. The critical value of α is determined in the infinite-volume limit by the condition

$$W'(\alpha)/V = \langle h(x) \rangle = 0, \tag{8.10}$$

where $H = \Sigma_x h(x)$. Thus α_{cr} is determined precisely by the condition that the expectation value of the BRS non-invariant term in the lagrangian density vanish. Moreover, in the naive continuum limit, α is the only scale-breaking parameter in the lagrangian, so the divergence of the dilation current $J_\mu = x_\nu T_{\mu\nu}$ is given by $\partial \cdot J = T_{\mu\mu} = 4\alpha h(x)$, apart from the anomaly. Here the energy–momentum tensor $T_{\mu\nu}$ is defined as in refs. [8,11], and has the property that its trace comes from the dimensionful parameters in the lagrangian density. Thus at the tree level the horizon condition gives $\langle T_{\mu\mu} \rangle = 0$. Consequently the BRS non-invariant and dilatation non-invariant term αH in the action does not lead to a breaking of dilatation invariance and does not contribute to the pressure at the tree level. Nevertheless, α sets the scale for mass in a gauge theory and ultimately contributes to $\langle T_{\mu\mu} \rangle \neq 0$ through the anomaly.

9. Effective action in lattice gauge theory

According to the scheme developed above, calculations are performed in two steps. Step 1: Evaluate expectation values, including $\langle h(x) \rangle$, with α an arbitrary parameter. Step 2: Determine the critical value α_{cr} from the horizon condition $\langle h(x) \rangle = 0$. Step 1 may be affected in a perturbatively renormalizable, weak-coupling expansion in powers of g_0 that is outlined in this section and in Appendix H. In fact this is the only practical calculational scheme in existence at the moment. On the other hand, step 2 is inherently non-perturbative. To see why this is so, recall that the horizon condition comes from $W'(\alpha) = 0$ (where $W = \ln Z$, and Z is the partition function), and this equation has a solution only if $W''(\alpha) \neq 0$, where $W''(\alpha)$ is the variance of H. However, the variance is a measure of quantum fluctuations, and therefore it vanishes in the tree approximation, $W''(\alpha) = 0$. The critical value, α_{cr}, is that value of α for which the tree approximation to $W'(\alpha)$ is exactly cancelled by the quantum corrections. Clearly one must include at least the leading quantum corrections to $W'(\alpha)$ to get an approximation which has a chance of being qualitatively correct. Thus we have an inherently non-perturbative calculational scheme (step 2), which is nevertheless consistent with perturbative renormalization (step 1).

When step 1 is evaluated perturbatively, the scheme is completely local because the only non-locality which is present, namely the cutoff at the Gribov horizon $\partial\Omega$, does not affect the perturbative calculation of expectation values. In the present section we shall show that this non-locality may also be eliminated non-perturbatively by means of the effective action Γ of the lattice theory. This quantity is determined by Schwinger–Dyson equations that are local, and satisfies a local Slavnov identity.

In order to define the effective action Γ on the lattice, it is necessary to introduce sources for some set of coordinates that parametrize the field variables. However, a complication arises because the fundamental fields of lattice gauge theory, the transporters $U_\mu(x)$ along a link, take values in a non-linear space that does not admit a global coordinatization, namely the group manifold of the SU(3) or, more generally, the SU(n) group. This is sometimes dealt with by using the exponential mapping, $U_\mu(x) = \exp[g_0 A_{0\mu}(x)]$, and expanding systematically and perturbatively in powers of g_0. However, at the non-perturbative level, the significance of this expansion is obscure, and the present work would be incomplete if it relied on this expansion, thereby ignoring some possible non-perturbative effects, while at the same time implementing the cutoff at the boundary of the fundamental modular region, which is non-perturbative.

Instead of the exponential mapping, we shall introduce another set of variables that allow a non-perturbative but yet purely local characterization of the effective action. These variables may also offer computational advantages in perturbative lattice calculations, as described in Appendix H.

Every term in $S_{\rm loc}$ is linear in the components of the $U_\mu(x)$, apart from the Wilson action $S_{\rm W}(U)$ which is quartic in these components. This suggests treating all the components of the $U_\mu(x)$ as independent linear variables, and introducing Lagrange-multiplier fields which serve to restrict these components to the group manifold. The effective action may then be defined by introducing sources for these fields. A similar approach may be found in ref. [12].

We consider here the especially simple case of the SU(2) group, and treat the SU(n) group for $n > 2$ in Appendix G. Consider the 2×2 matrix

$$b = b^0 - i \sum_{a=1}^{3} \sigma^a b^a, \qquad (9.1)$$

where b^a is an arbitrary real four-vector. The linear mapping $b \to b' = g_1 b g_2^\dagger$, where g_1 and g_2 are elements of the SU(2) group, induces an O(4) rotation on b^a. Consequently the measure

$$\int \mathrm{d}^4 b\ \delta\big(|b|^2 - 1\big) = \int \mathrm{d}^4 b\ \mathrm{d}\mu\ \exp\big[\mu\big(|b|^2 - 1\big)\big], \tag{9.2}$$

where $|b|^2 \equiv \Sigma_{a=0}^3 (b^a)^2$, is invariant under left and right multiplication by g, and is supported on the group manifold of the SU(2) group, namely the three-sphere $|b|^2 = 1$. Consequently it represents the Haar measure. The five-dimensional volume element $\mathrm{d}^4 b\ \mathrm{d}\mu$ is cartesian, and the auxiliary variable μ acts like a Lagrange multiplier to enforce the constraint $|b|^2 = 1$. The integral over μ runs along the imaginary axis.

This representation of the Haar measure is written for each link (xy). Thus a non-unitary matrix b_{xy}, given by eq. (9.1), and a Lagrange-multiplier field μ_{xy} are assigned to the link (xy), with $b_{yx} = (b_{xy})^\dagger \neq (b_{xy})^{-1}$, and $\mu_{yx} = \mu_{yx}$. The volume element is given by $\Pi_{(xy)}\, \mathrm{d}^4 b_{xy}\, \mathrm{d}\mu_{xy}$. For each link (xy), the exponent $\mu_{xy}(|b_{xy}|^2 - 1)$ is added to the action, which in total gets modified by the addition of the constraint action

$$S_{\mathrm{cs}}(b, \mu) \equiv \big(\mu, |b|^2 - 1\big) \equiv \sum_{(xy)} \mu_{xy}\big(|b_{xy}|^2 - 1\big). \tag{9.3}$$

There are now five independent elementary fields $b_\nu^a(x)$ and $\mu_\nu(x)$ for each link (x, ν). The gain is that the volume element is cartesian and the lattice action is polynomial and local in these variables. Moreover, invariance under the local gauge group,

$$b_{xy} \to g_x b_{xy} g_y^\dagger, \quad \mu_{xy} \to \mu_{xy}. \tag{9.4}$$

is preserved, because the volume element $\Pi\, \mathrm{d}b\, \mathrm{d}\mu$ and the constraint action $S_{\mathrm{cs}}(b, \mu)$ are invariant under this transformation, as we have seen. The Wilson action $S_{\mathrm{W}}(b)$ is also invariant, even though the b_{xy} are not unitary. In the absence of gauge-fixing, the total gauge-invariant action is given by

$$S_{\mathrm{gi}} \equiv S_{\mathrm{cs}}(b, \mu) + \beta S_{\mathrm{W}}(b). \tag{9.5}$$

A locally gauge-invariant generating functional $Z(J, m)$ may be defined by introducing five sources for each link, namely, a real four-vector $(J_{xy})^a$ for the four independent components $(b_{xy})^a$, and a real source m_{xy} for μ_{xy},

$$Z(J, m) = \int \mathrm{d}b\ \mathrm{d}\mu\ \exp\big[-S_{\mathrm{gi}} + (J, b) + (m, \mu)\big], \tag{9.6}$$

where

$$(J, b) + (m, \mu) \equiv \sum_{x,\nu} [J_\nu^a(x) b_\nu^a(x) + m_\nu(x)\mu_\nu(x)]. \tag{9.7}$$

A locally gauge-invariant effective action $\Gamma(b, \mu)$ is obtained by Legendre transformation from $W(J, m) \equiv \ln[Z(J, m)]$. Although this effective action is well defined, it does not possess a weak-coupling expansion, and for this reason we return to the gauge-fixed action, obtained by integrating over the fundamental modular region.

In this case the total action contains the additional terms previously found,

$$S_{\text{phys}} \equiv S_{\text{gi}} + s\psi + \alpha H_{\text{loc}}, \tag{9.8}$$

where ψ is given in eq. (8.4b), and H_{loc} in eqs. (6.13) and (6.14). The explicit form of S_{phys} is given below, eq. (9.12). The action of the BRS operator s on the fields $b_\nu(x)$ and $\mu_\nu(x)$ follows from their transformation under local gauge transformation, so the action of s on the matrix $b_\nu(x)$ retains its previous form, e.q (8.2), even though $b_\nu(x)$ is not unitary, and it leaves the new link variable $\mu_\nu(x)$ invariant,

$$sb_{xy} = -C_x b_{xy} + b_{xy} C_y, \tag{9.9a}$$

$$s\mu_\nu(x) = 0. \tag{9.9b}$$

The action of s on other fields is given in eq. (8.2). The partition function previously defined is given by

$$Z(\alpha) = \int_\Omega \mathrm{d}b\ \mathrm{d}\mu\ \mathrm{d}\lambda\ \mathrm{d}C\ \mathrm{d}C^*\ \mathrm{d}\varphi\ \mathrm{d}\bar{\varphi}\ \mathrm{d}\omega\ \mathrm{d}\bar{\omega}\ \exp(-S_{\text{phys}}), \tag{9.10}$$

and α is determined by the horizon condition $W'(\alpha)/V = 0$, where $W(\alpha) = \ln\ Z(\alpha)$.

The partition function gets promoted to the generating function of correlation functions by the addition of sources to the action. As always in the BRS method, we introduce local sources for the composite BRS-transform fields, namely $K_\nu^a(x)$, with $a = 0, \ldots, 3$, for $sb_\nu^a(x)$, and $L^a(x)$ for $sC^a(x)$,

$$(K, sb) + (L, sC) \equiv \sum_x K_\nu^a(x) sb_\nu^a(x) + L^a(x) sC^a(x). \tag{9.11}$$

To prove renormalizability for all the terms that appear in S_{phys}, it is convenient to introduce a local source $M_{\mu i}^a(x)$ for each component of $[D_\mu(b)\varphi_i]^a(x)$, and similarly a source $N_{\mu i}^a(x)$ for $[D_\mu(b)\bar{\omega}_i]^a(x)$, following the procedure for continuum gauge theory in ref. [8]. (Note however that the contour of integration chosen for the ghost field $\bar{\varphi}$ is real in ref. [8] and imaginary here.) We have written the single index $i \equiv (\nu, b)$ for the pair of mute indices, so i takes on $f = (n^2 - 1)D$ values.

We also introduce sources $W_{\mu i}^a(x)$ and $V_{\mu i}^a(x)$ for the BRS transforms of these quantities, $s[D_\mu(b)\varphi_i]^a(x)$ and $s[D_\mu(b)\overline{\omega}_i]^a(x)$.

The explicit expression for S_{phys} is given by

$$S_{\text{phys}} = \left(\mu, |b|^2 - 1\right) + \beta S_{\text{W}}(b) + (\lambda, \nabla \cdot A) + (C^*, M(b)C)$$
$$-(\overline{\varphi}, M(b)\varphi) + (\overline{\omega}, M(b)\omega) + (\overline{\omega}, M(sb)\varphi)$$
$$-\alpha^{1/2}(D(b)\varphi) - \alpha^{1/2}s(D(b)\overline{\omega}) - \alpha(n^2-1)E(b), \qquad (9.12)$$

where $E(b)$ is given in eq. (1.2b), and $(D(b)\varphi) = \Sigma_{x,\mu,c}[D_\mu(b)\varphi]_{\mu c}^c(x)$, and similarly for $(D(b)\overline{\varphi})$. (This expression comes from the action (6.4b) rather than (6.4a).) The extended action in the presence of all composite sources is defined by

$$-S \equiv -S_{\text{phys}} + (K, sb) + (L, sC) + (M, (D(b)\varphi)) + (N, (D(b)\overline{\omega}))$$
$$+(W, s(D(b)\varphi)) + (V, s(D(b)\overline{\omega})). \qquad (9.13)$$

The action with the sources for all elementary fields is given by

$$-\Sigma = -S + (J, b) + (m, \mu) + (\eta^*, C) + (C^*, \eta) + (l, \lambda)$$
$$+(\rho^*, \varphi) + (\overline{\varphi}, \rho) + (\sigma^*, \omega) + (\overline{\omega}, \sigma). \qquad (9.14)$$

Note that there are local sources $M(x)$ and $V(x)$ in eq. (9.13) for each of the composite fields $(D(b)\varphi)(x)$ and $s(D(b)\overline{\omega})(x)$ that have coefficient $\alpha^{1/2}$ in the action S_{phys} and a local source $J_\mu^0(x)$ in eq. (9.14) for the elementary field n^{-1} Re tr$[b_\mu(x)]$ that has coefficient α in S_{phys}, and also for the BRS transforms of these quantities.

The generating function of correlation functions is given by

$$Z(\boldsymbol{J}, \boldsymbol{M}, \alpha) = \exp[W(\boldsymbol{J}, \boldsymbol{M}, \alpha)] = \int_\Omega \mathrm{d}\Phi \exp(-\Sigma). \qquad (9.15)$$

Here $\mathrm{d}\Phi$ represents the volume element for all bosonic and fermionic fields, $\boldsymbol{J}$ represents the set of all elementary sources, and $\boldsymbol{K}$ represents the set of sources for all composite fields,

$$\boldsymbol{J} \equiv (J, m, \eta, \eta^*, l, \rho, \rho^*, \sigma, \sigma^*), \qquad (9.16)$$

$$\boldsymbol{K} \equiv (K, L, M, N, W, V). \qquad (9.17)$$

The Schwinger–Dyson equations for $Z(\boldsymbol{J}, \boldsymbol{M}, \alpha)$ are derived from the identity

$$0 = \int_{\Omega} \mathrm{d}\Phi \frac{\partial \exp(-\Sigma)}{\partial \Phi_i}, \tag{9.18}$$

where Φ_i represents a generic bosonic or fermionic field variable. There is no contribution from the boundary because, as noted previously, the Boltzmann factor $\exp[-\alpha H(b)]$ and $\det[M(b)]$ vanish on $\partial\Omega_L$. The Schwinger–Dyson equations for the partition function Z have the standard form

$$S_{,i}\left(\frac{\partial}{\partial \boldsymbol{J}}\right) Z(\boldsymbol{J}, \boldsymbol{M}, \alpha) = \boldsymbol{J}_i Z(\boldsymbol{J}, \boldsymbol{M}, \alpha), \tag{9.19}$$

where $S_{,i}(\Phi) \equiv \partial S(\Phi)/\partial \Phi_i$, and $\boldsymbol{J}_i$ is the source of the generic field component Φ_i. Here S is the local extended action that is given explicitly in eqs. (9.12) and (9.13).

The Legendre transformation from $W(\boldsymbol{J}, \boldsymbol{K}, \alpha) = \ln Z(\boldsymbol{J}, \boldsymbol{K}, \alpha)$ to the effective action $\Gamma(\Phi, \boldsymbol{K}, \alpha)$ is given by

$$\Gamma(\Phi, \boldsymbol{K}, \alpha) = \boldsymbol{J}_i \Phi_i - W(\boldsymbol{J}, \boldsymbol{K}, \alpha). \tag{9.20}$$

The Schwinger–Dyson equation for the effective action follows in the standard way,

$$\frac{\partial \Gamma(\Phi, \boldsymbol{K}, \alpha)}{\partial \Phi_i} = S_{,i}\left(\Phi_j + D_{jk} \frac{\partial}{\partial \Phi_k}\right) 1, \tag{9.21}$$

where the propagator $D_{ij} = D_{ij}(\Phi, \boldsymbol{M})$ is inverse to the matrix $\partial^2 \Gamma(\Phi, \boldsymbol{K}, \alpha)/\partial\Phi_i\, \partial\Phi_j$. Correlation functions are obtained by differentiation with respect to the sources, after which they are assigned the physical values $\boldsymbol{J} = 0$ and $\boldsymbol{K} = 0$. For the effective action, this corresponds to values Φ_{phys} determined as usual by

$$\frac{\partial \Gamma(\Phi_{\mathrm{phys}}, \boldsymbol{0}, \alpha)}{\partial \Phi_i} = 0. \tag{9.22}$$

The extended action (9.12) contains local sources for the BRS transforms of the BRS-violating terms in the action, namely $\Sigma_{x,\mu}$ Re $\mathrm{tr}[b_\mu(x)]$ and $\Sigma_{x,\mu} D_\mu^{ac}(b)\varphi_{\mu a}^{c}(x)$ so that lattice Slavnov–Taylor identities may be derived. They have the standard local form,

$$\sum_{x,a,\mu} \frac{\partial \Gamma}{\partial b_\mu^a(x)} \frac{\partial \Gamma}{\partial K_\mu^a(x)} + \sum_{x,a} \frac{\partial \Gamma}{\partial C^a(x)} \frac{\partial \Gamma}{\partial L^a(x)} + \ldots = 0. \tag{9.23}$$

Thus the lattice provides a non-perturbative and BRS-covariant regularization.

Finally the thermodynamic parameter $\alpha = \alpha(g)$ is determined by the horizon condition which takes the simple form

$$\frac{\partial \Gamma(\Phi_{\text{phys}}, \mathbf{0}, \alpha)}{\partial \alpha} = -\frac{\partial W(\mathbf{0}, \mathbf{0}, \alpha)}{\partial \alpha} = 0. \tag{9.24}$$

Eqs. (9.21)–(9.24) provide a complete and local description of the lattice gauge theory at infinite volume, obtained after integration over the fundamental modular region. In Appendix H we indicate how a lattice perturbation theory based on these equations may be developed. Moreover, it has been proven, using the Slavnov–Taylor identities (9.24) that these solutions are perturbatively renormalizable [8], so that the effective action satisfies

$$\Gamma(\Phi, \boldsymbol{K}, \alpha, g) = \Gamma_{\text{ren}}(\Phi_{\text{ren}}, \boldsymbol{K}_{\text{ren}}, \alpha_{\text{ren}}, g_{\text{ren}}). \tag{9.25}$$

These renormalizations require precisely two independent renormalization constants, just as in Faddeev–Popov theory in the Landau gauge. It was also proven [8] that the horizon condition is a finite equation in terms of renormalized quantities.

10. Dipole in the Faddeev–Popov propagator

Consider the propagator corresponding to the Faddeev–Popov matrix $M = -\nabla \cdot D(U)$,

$$G(x-y)\delta^{ab} \equiv \langle M^{-1\,ab}(x, y) \rangle = \langle C^a(x) C^{*b}(y) \rangle, \tag{10.1}$$

which is also the Fermi-ghost propagator. In this section we shall show that the horizon condition is equivalent to the condition that the Fourier transform of this propagator,

$$G(\theta) \equiv \sum_x G(x) \exp(-i\theta \cdot x), \tag{10.2}$$

possesses a dipole ghost singularity at $\theta = 0$ given by

$$G(\theta) \sim c/\lambda^2(\theta), \tag{10.3}$$

where $c > 0$, θ_μ is the lattice Brillouin momentum, $-\pi < \theta_\mu \leqslant \pi$, and

$$\lambda(\theta) \equiv \sum_\mu 4 \sin^2\left(\tfrac{1}{2}\theta_\mu\right) \tag{10.4}$$

is the eigenvalue of the lattice laplacian, $(-\nabla \cdot \nabla) \exp(i\theta \cdot x) = \lambda(\theta) \exp(i\theta \cdot x)$.

The presence of a dipole singularity was used by Gribov [2], in a second-order continuum calculation, as a condition to define the boundary $\partial\Omega$ of the region where the Faddeev–Popov operator M is positive (called here the Gribov region Ω). It was obtained in ref. [8] from the geometrical definition of this region, to all orders in continuum theory by a functional method. We shall show here for the lattice theory, by the bubble-diagram expansion of the ghost self-energy, that it is a consequence of the cutoff at the boundary of the augmented core region Ψ, which is a proper subset of the Gribov region, $\Psi \subset \Omega$.

We first express the horizon condition as a condition on the related tensor propagator,

$$G_{\mu\nu}(x-y)\delta^{ab} \equiv \left\langle D_{\mu}^{ac}(x) D_{\nu}^{bd}(y) M^{-1cd}(x, y) \right\rangle. \tag{10.5}$$

Its Fourier transform,

$$G_{\mu\nu}(\theta) \equiv \sum_{x} G_{\mu\nu}(x) \exp\left[-i\theta \cdot \left(x + \tfrac{1}{2}e_{\mu} - \tfrac{1}{2}e_{\nu}\right)\right], \tag{10.6}$$

is closely related to the expectation value of the horizon function $H(U)$, eq. (4.4a),

$$\langle H \rangle / V = \left[\lim_{\theta \to 0} G_{\mu\mu}(\theta) - e \right] (n^2 - 1), \tag{10.7}$$

where

$$e \equiv \langle E(U) \rangle / V \tag{10.8}$$

and $E(U)$ is the maximizing function, eq. (1.2b), that is proportional to the euclidean volume V. The horizon condition at infinite volume, eq. (8.10), reads

$$\lim_{\theta \to 0} \sum_{\mu} G_{\mu\mu}(\theta) = e. \tag{10.9}$$

Here the limit $\theta \to 0$ provides an infrared regularization which will be needed, because it turns out that the strictly zero-frequency Fourier component is ambiguous in the infinite-volume limit.

Observe from eq. (10.5) that $G_{\mu\nu}(x-y)$ satisfies

$$\begin{aligned}
\nabla^{(x)} \cdot G_{\nu}(x-y)\delta^{ab} &= -\left\langle D_{\nu}^{ba}(y)\delta_{x,y} \right\rangle \\
&= -\left\langle G_{\nu}^{ba}(y) \right\rangle \left(\delta_{x,y+\nu} - \delta_{x,y}\right) \\
&= -\delta^{ba}\left(\delta_{x,y+\nu} - \delta_{x,y}\right) e/D,
\end{aligned}$$

where we have used the definition of the gauge-covariant derivative, eqs. (B.11) and (B.12c), and the equation $\langle A_\nu^b(y)\rangle = 0$, which holds by lattice or global color symmetry. In momentum space this reads

$$\sum_\mu q_\mu(\theta) G_{\mu\nu}(\theta) = q_\nu(\theta) e/D, \tag{10.10}$$

where we have introduced the lattice "momentum" vector

$$q_\mu(\theta) \equiv 2\sin\left(\tfrac{1}{2}\theta_\mu\right), \tag{10.11}$$

that satisfies

$$q^2(\theta) = \lambda(\theta). \tag{10.12}$$

The tensor propagator $G_{\mu\nu}(\theta)$ is thus of the form

$$G_{\mu\nu}(\theta) = G_{\mu\nu}^{\mathrm{TT}}(\theta) + e q_\mu(\theta) q_\nu(\theta)\left[D q^2(\theta)\right]^{-1}, \tag{10.13}$$

where $G_{\mu\nu}^{\mathrm{TT}}(\theta)$ is transverse in both indices, $q_\mu(\theta) G_{\mu\nu}^{\mathrm{TT}}(\theta) = G_{\mu\nu}^{\mathrm{TT}}(\theta) q_\nu(\theta) = 0$. This allows us to express the horizon condition (10.9) as a condition on $G_{\mu\nu}^{\mathrm{TT}}(\theta)$,

$$\lim_{\theta\to 0} \sum_\mu G_{\mu\mu}^{\mathrm{TT}}(\theta) = eD^{-1}(D-1). \tag{10.14a}$$

Note that the value provided by the horizon condition is the only one which makes the tensor $G_{\mu\nu}(\theta)$ regular at $\theta = 0$,

$$\lim_{\theta\to 0} G_{\mu\nu}(\theta) = eD^{-1}\delta_{\mu\nu}. \tag{10.14b}$$

We wish to express the horizon condition as a condition on the propagator $G(\theta)$ of the Faddeev–Popov operator. For this purpose we separate out the lattice laplacian from the Faddeev–Popov operator,

$$M = -\nabla\cdot D = -\nabla^2 + M_1 = M_0 + M_1, \tag{10.15}$$

and write, in operator notation,

$$M^{-1} = M_0^{-1} - M_0^{-1} M_1 M_0^{-1} + M_0^{-1} M_1 M^{-1} M_1 M_0^{-1}, \tag{10.16}$$

which gives

$$\langle M^{-1}\rangle = M_0^{-1} + M_0^{-1}\langle -M_1 + M_1 M^{-1} M_1\rangle M_0^{-1}. \tag{10.17}$$

In momentum space this reads

$$G(\theta) = \left[q^2(\theta)\right]^{-1} + \left[q^2(\theta)\right]^{-1} q_\mu(\theta) K_{\mu\nu}(\theta) q_\nu(\theta) \left[q^2(\theta)\right]^{-1}, \quad (10.18)$$

where tensor $K_{\mu\nu}(\theta)$ is defined by

$$K_{\mu\nu}(\theta)\delta^{ab} \equiv \left(\delta^{ab} - \left\langle G_\mu^{ab} \right\rangle\right)\delta_{\mu\nu} + L_{\mu\nu}(\theta)\delta^{ab},$$

$$K_{\mu\nu}(\theta) = \left[1 - \langle e(U) \rangle / D\right]\delta_{\mu\nu} + L_{\mu\nu}(\theta). \quad (10.19)$$

Here $L_{\mu\nu}(\theta)$ is the Fourier transform of the propagator

$$L_{\mu\nu}(x-y)\delta^{ab} \equiv \left\langle V_\mu^{ac}(x) V_\nu^{bd}(y) M^{-1cd}(x, y) \right\rangle, \quad (10.20)$$

where $V_\mu^{ac} \equiv (D_\mu^{ac} - \nabla_\mu \delta^{ac})$. To obtain this expression, we have again used eq. (B.12b), and $\langle A_\nu^b(y) \rangle = 0$. The form of $M_1 = -\nabla \cdot (D - \nabla) = -(D - \nabla) \cdot \nabla$, allowed us to make explicit the factors $q_\mu(\theta)$ and $q_\nu(\theta)$. Observe that the part of $L_{\mu\nu}(\theta)$ which is transverse in both indices is given by

$$L_{\mu\nu}^{\mathrm{TT}}(\theta) = G_{\mu\nu}^{\mathrm{TT}}(\theta), \quad (10.21)$$

and consequently the horizon condition, eq. (10.14a), reads

$$\lim_{\theta \to 0} \sum_\mu K_{\mu\mu}^{\mathrm{TT}}(\theta) = D - 1. \quad (10.22)$$

The well-known bubble-diagram expansion for the Fermi-ghost self-energy gives

$$\left[G(\theta)\right]^{-1} = q^2(\theta) - q_\mu(\theta) K_{\mathrm{I}\mu\nu}(\theta) q_\nu(\theta), \quad (10.23)$$

where $K_{\mathrm{I}\mu\nu}(\theta)$ is the sum of the one-particle-irreducible diagrams that contribute to $K_{\mu\nu}(\theta)$,

$$K_{\mu\nu}(\theta) = K_{\mathrm{I}\mu\nu}(\theta) + K_{\mathrm{I}\mu\kappa}(\theta) q_\kappa(\theta) \left[q^2(\theta)\right]^{-1} q_\lambda(\theta) K_{\mathrm{I}\lambda\nu}(\theta) + \dots \, . \quad (10.24)$$

The factorization of the ghost "momentum" $q_\kappa(\theta)$ is again explicit. Because $K_{\mathrm{I}\mu\nu}(\theta)$ is a sum of one-particle-irreducible diagrams, it is a regular function of $q_\lambda(\theta)$ at $\theta = 0$ in every order of perturbation theory. In the absence of (additional) non-perturbative effects, this is an exact property of the theory. Consequently, by lattice symmetries, $K_{\mathrm{I}\mu\nu}(\theta)$ has the expansion about $\theta = 0$ given by

$$K_{\mathrm{I}\mu\nu}(\theta) = a\delta_{\mu\nu} + bq^2(\theta)\delta_{\mu\nu} + cq_\mu(\theta) q_\nu(\theta) + \dots \, . \quad (10.25)$$

(Terms that break euclidean invariance are of higher order in θ.) From the last two equations, we have

$$\lim_{\theta \to 0} K_{\mu\nu}^{\mathrm{TT}}(\theta) = \lim_{\theta \to 0} K_{\mathrm{I}\mu\nu}^{\mathrm{TT}}(\theta), \tag{10.26}$$

and the horizon condition (10.22) reads

$$\lim_{\theta \to 0} \sum_{\mu} K_{\mathrm{I}\mu\mu}^{\mathrm{TT}}(\theta) = D - 1, \tag{10.27}$$

which gives

$$K_{\mathrm{I}\mu\nu}(\theta) = \delta_{\mu\nu} + bq^2(\theta)\delta_{\mu\nu} + cq_\mu(\theta)q_\nu(\theta) + \dots \,. \tag{10.28}$$

When this is substituted into eq. (10.23), we obtain

$$[G(\theta)]^{-1} = \mathrm{O}\left\{[q^2(\theta)]^2\right\}. \tag{10.29}$$

The horizon condition is thus the statement that the higher order quantum contributions to the Fermi-ghost self-energy precisely cancel the zero-order self-energy at $\theta = 0$! Moreover, c is positive because the lattice Faddeev–Popov operator M is positive in the Gribov region Ω, on which the measure is supported. QED.

Remark 1. We wish to emphasize that this result is dependent on the hypothesis that the one-particle-irreducible self-energy part $K_{\mathrm{I}\mu\nu}(\theta)$ is regular at $\theta = 0$.

Remark 2. The horizon condition may alternatively be expressed as

$$\lim_{\theta \to 0} \left[G(\theta)^{-1}/\lambda(\theta)\right] = 0, \tag{10.30a}$$

which gives

$$\lim_{\theta \to 0} \left[G_{\mathrm{ren}}(\theta)^{-1}/\lambda(\theta)\right] = 0, \tag{10.30b}$$

where $G_{\mathrm{ren}}(\theta) = G(\theta)/Z$ is the renormalized propagator. This formulation of the horizon condition makes it obvious that the horizon condition gives a finite relation between renormalized quantities, provided that the renormalized Faddeev–Popov propagator retains the same infrared behavior as the unrenormalized propagator when the ultraviolet cutoff is removed.

Remark 3. It is not surprising that in the infinite-volume limit, entropy should drive the support of the probability distribution to the boundary $\partial\Psi$ of the augmented core region Ψ, or equivalently, to the boundary $\partial\Lambda$ of the fundamental modular region Λ. However, the dipole singularity of the Faddeev–Popov propagator shows that, more particularly, the probability distribution is strongly weighted on that part of $\partial\Psi$ which coincides with the boundary $\partial\Omega$ of the Gribov region Ω where M has a non-trivial vanishing eigenvalue. In fact, as discussed in ref. [8] for the continuum case, the dipole singularity of the lattice Faddeev–Popov propagator shows that $\langle\rho(\lambda)\rangle$, the expectation value of the density per unit volume of the eigenvalues of the Faddeev–Popov operator, diverges at $\lambda = 0$ as compared to the density of eigenvalues of the lattice laplacian at $\lambda = 0$. At first sight this appears paradoxical, because the augmented core region Ψ is defined by the condition $H(U) < 0$, whereas in general $H(U) \to \infty$ as the boundary $\partial\Omega$ of the Gribov region is approached. However, the paradox is resolved by the observation that the tensor propagator $G_{\mu\nu}(\theta)$, which is obtained from $G(\theta)$ by the application of two gauge-covariant derivatives $D_\mu D_\nu$ to M^{-1} is regular at $\theta = 0$, eq. (10.14b). This shows that the eigenfunctions ω belonging to (approximately) vanishing eigenvalues of M are (approximately) annihilated by the gauge-covariant derivative, $D_\mu\omega \approx 0$, and consequently they are generators of (approximately) degenerate gauge orbits. Such orbits are in fact intersection points of the boundary $\partial\Lambda$ of the fundamental modular region with the boundary $\partial\Omega$ of the Gribov region.

Remark 4. Recall that the theory with two independent parameters g and α represents lattice gauge theory only when the horizon condition holds, $\alpha = \alpha_{\mathrm{cr}}$ (and then, only at infinite volume). It is helpful to let α approach α_{cr} from above, so $\langle H\rangle < 0$, as it is on a finite lattice, and to regain the horizon condition in the limit $\alpha \to \alpha_{\mathrm{cr}}$. In this limit, the Faddeev–Popov propagator develops a dipole singularity in the infrared, which is not seen in any finite order of perturbation theory. This signals the onset of non-perturbative infrared divergences. It is to be expected that other correlation functions, the gluon propagator among them, change qualitatively in this limit. This is welcome, because the gluon propagator in finite order of weak-coupling perturbation theory contains non-physical singularities. However, it means that perturbation theory is not a reliable guide to the behavior of these correlation functions. For this reason, it will be helpful to review what non-perturbative information we have about the gluon propagator at low momenta.

Remark 5. We first consider what we may learn from the results of the present section. As noted above, eq. (10.14b), the tensor quantity $G_{\mu\nu}(\theta)$, which contains the lattice gauge-covariant derivatives D_μ and D_ν, is regular at $\theta = 0$ when the horizon condition holds. On the other hand $J_{\mu\nu}(\theta)$,

$$J_{\mu\nu}(x-y)\delta^{ab} \equiv \left\langle \nabla_\mu(x)\nabla_\nu(y)M^{-1\,ab}(x, y)\right\rangle, \tag{10.31}$$

which is obtained from $G_{\mu\nu}(\theta)$ by replacing the lattice gauge-covariant derivative by the ordinary lattice difference, has the Fourier transform

$$\begin{aligned} J_{\mu\nu}(\theta) &= \sum_x J_{\mu\nu}(x)\exp\left[-i\theta\cdot\left(x + \tfrac{1}{2}e_\mu - \tfrac{1}{2}e_\nu\right)\right] \\ &= q_\mu(\theta)q_\nu(\theta)G(\theta), \end{aligned} \tag{10.32}$$

and the singularity $q_\mu(\theta)q_\nu(\theta)[q^2(\theta)]^{-2}$ at $\theta=0$. Thus $J_{\mu\nu}(\theta)$ is highly singular at $\theta=0$. This implies that when the lattice gauge covariant derivative D appears in the horizon function, there is a strong cancellation between the difference and the connection pieces. (In continuum form, $D=\partial+A$.) Furthermore, because $G_{\mu\nu}(\theta)$ is regular at $\theta=0$, it follows that $L_{\mu\nu}(\theta)$, eq. (10.20), which contains the quantity $V_\mu^{ac}=(D_\mu-\nabla_\mu)^{ac}$, that becomes the connection $f^{abc}A_\mu^b$ in the continuum limit, has the same singularity as $J_{\mu\nu}(\theta)$.

On the other hand, $J_{\mu\nu}(\theta)$ formally vanishes at strictly zero frequency, $J_{\mu\nu}(0)=0$, because it is the sum of lattice differences,

$$J_{\mu\nu}(0)=\sum_x \nabla_\mu J_\nu(x)=\sum_x\left[J_\nu(x+e_\mu)-J_\nu(x)\right], \tag{10.33}$$

where

$$J_\nu(x)=\left|\left\langle \nabla_\nu(y)M^{-1\,ab}(x,y)\right\rangle\right|_{y=0}. \tag{10.34}$$

This shows the necessity of using θ as an infrared regularizer, and then taking the limit $\theta\to 0$, as was done in evaluating the horizon condition in this section. To be consistent, the same infrared regularization of the horizon function must also be implemented where the horizon function appears in the action. Thus we should take $\exp[i\theta\cdot(x-y)]D_\mu^{(x)}D_\mu^{(y)}M^{-1}(x,y)$ in the non-local form of the action, and let θ approach zero after the thermodynamic parameter α approaches α_{cr}, or use the local source $\exp(i\theta\cdot x)$ for the BRS-violating term in the local action, as discussed in sect. 8 and in ref. [8]. In perturbative calculations [2,5,7], no infrared regulator is used, which means that $J_{\mu\nu}(0)$ is implicitly assigned its formal value $J_{\mu\nu}(0)=0$. This neglects the derivative pieces in $G_{\mu\nu}(0)$, and thus also the cancellation in $D=\partial+A$ between the derivative and the connection pieces which, we have seen, occurs in the horizon function at α_{cr}. Consequently in these perturbative calculations, the infrared part of the connection A may be too strongly suppressed, so that the resulting gluon propagator $D(k)$, which vanishes like k^2 at $k=0$ in every order of perturbation theory, is likely to be too small at zero frequency.

Remark 6. Some exact results on the infrared behavior of the lattice gluon propagator in the minimal Landau gauge may be found in ref. [6]. In particular, for SU(n) in d euclidean dimensions, and for any probability distribution with support in the Gribov region Ω, the bound (3.5a) of ref. [6] holds,

$$(2\pi)^{-d}\int \mathrm{d}^d\theta\; D(\theta)\tau(\theta)/\lambda(\theta)<4dn, \tag{10.35}$$

where $\tau(0)=1$. Consequently, in dimension $d=4$, the lattice gluon propagator $D(\theta)$ cannot be as singular as a simple gluon pole $\lambda^{-1}(\theta)$ at $\theta=0$.

We also review another result of ref. [6], where the partition function $Z(J) = \exp[W(J)]$, is considered for the particular form of the source $J_\mu^a(x) = H \cos(\theta \cdot x)\delta_{\mu 1}\delta^{a3}$. By analogy with lattice spin models, H may be viewed as the strength of a spatially modulated magnetic field of wave number θ. Thus, with $W(J) \equiv W(H, \theta) = w(H, \theta)V$, where V is the euclidean volume, the gluon propagator $D(H, \theta)$, in the presence of the source parametrized by H, is given by $D(H, \theta) = \partial^2 w(H, \theta)/\partial H^2$. This includes, as a special case, the standard gluon propagator $D(\theta) = D(0, \theta)$. For any probability distribution with support in the Gribov region Ω, the bound (4.8) of ref. [6] holds in the infinite-volume limit, namely, $w(H, \theta) \leqslant \sigma(\theta)H$, where $\sigma(\theta)$ behaves like $(2\lambda)^{1/2}(\theta)$ for small θ, so $\lim_{\theta \to 0} \sigma(\theta) = 0$, and $\lim_{\theta \to 0} w(H, \theta) = 0$. We also have $w(0, \theta) = 0$ and $\partial w(0, \theta)/\partial H = 0$, which gives $w(H, \theta) = \int_0^H \mathrm{d}y\,(H-y)D(y, \theta)$. Because the propagator is positive, $D(H, \theta) \geqslant 0$, we obtain $\lim_{\theta \to 0} D(H, \theta) = 0$, for almost every H. This is indeed a remarkable suppression of the propagator $D(H, \theta)$ in the infrared. With the additional hypothesis of ref. [6], that $D(H, \theta)$ is regular in H, it follows that $\lim_{\theta \to 0} D(\theta) = \lim_{\theta \to 0} D(0, \theta) = 0$, namely that the gluon propagator vanishes at $\theta = 0$. However this hypothesis has not been proven, and because "almost every" is not the same as "every", the behavior of $D(\theta)$ at $\theta = 0$, is not yet settled. (I am indebted to Pierre van Baal for a helpful correspondence on this point.)

From the above discussion we may conclude however that in four euclidean dimensions the gluon propagator $D(\theta)$ is less singular at $\theta = 0$ than $\lambda^{-1}(\theta)$ but, very likely, does not vanish as rapidly as $\lambda(\theta)$. One interesting possibility is that $D(\theta)$ is finite at $\theta = 0$. Indeed, this hypothesis appears to be consistent with recent numerical investigations of the gauge-fixed propagator [13]. For, according to fig. 3 of ref. [13], the gluon propagator at zero four-momentum $D(0)$ has the same numerical value on two different lattice volumes, whereas it should exhibit a volume dependence if it diverges or vanishes in the limit of infinite lattice volume. However, it should be noted that the algorithm for gauge-fixing, which numerically seeks a minimum of the minimizing function (1.3), may lead to a local instead of an absolute minimum, and it is not known what effect this may have on the expectation value of a gauge-dependent quantity such as the gluon propagator.

Remark 7. It was noted above that the horizon condition is equivalent to the regularity of the tensor $G_{\mu\nu}(\theta)$ at $\theta = 0$. It is remarkable that the continuum analog of this condition agrees precisely with the confinement criterion of Kugo and Ojima [14].

11. Area law

In the preceding sections, we have formulated lattice gauge theory in the infinite-volume limit in the minimal Landau gauge. With the introduction of auxiliary or ghost fields, this gauge-fixed theory may be described by a local action that depends upon two independent coupling parameters g and α, which however are related by a supplementary or "horizon" condition, $\langle H \rangle = 0$. In the last section, we found that the horizon condition is equivalent to the condition that the lattice Faddeev–Popov propagator $\langle M^{-1} \rangle$ develops a dipole singularity at zero momentum, which indicates the presences of very long range spatial correlations. In the present concluding section, we shall show that these long range correlations are capable of causing confinement. The basic reason is that M^{-1} appears in the action αH.

Wilson proposed that a confining theory should exhibit an area law,

$$W = \langle \mathrm{tr}\, U_p \rangle \sim \exp(-KA), \tag{11.1}$$

where U_p is the transporter around a closed planar curve p, A is the area enclosed, and K is the string tension. Such an area law is found in strong-coupling expansions of lattice gauge theory. In the continuum limit, the thermodynamic parameter α, introduced in sect. 5, has dimensions of (mass)^4, so $\alpha^{1/2}$ has the same dimension as the string tension K. To obtain a well-defined continuum limit from the lattice theory, the horizon condition, $\langle H \rangle = 0$, should be satisfied by solving for $g = g_{\mathrm{cr}}(\alpha)$, instead of $\alpha = \alpha_{\mathrm{cr}}(g)$, so that the thermodynamic parameter α, which is dimensionful in the continuum limit, is the independent parameter instead of g. This constitutes a realization of dimensional transmutation. After renormalization of g and α, the equation $g = g_{\mathrm{cr}}(\alpha)$ is finite in the continuum limit. This is manifest from the alternative expression of the horizon condition obtained in the last section, namely, that the horizon condition may be expressed by $G^{-1}(k)/k^2|_{k=0} = 0$, where $G(k)$ may be either the unrenormalized or renormalized lattice Faddeev–Popov propagator. Moreover, this equation is consistent with the perturbative renormalization group [5,8].

(For the purpose of a weak-coupling expansion it is convenient to rescale α according to $\alpha = \gamma/g^2$. The renormalization-group equations for the unrenormalized quantities g and γ were solved in ref. [5,8], with the result

$$g^{-2} = 2b_0 \ln(\Lambda/\Lambda_{\mathrm{latt}}) + \dots,$$

$$\gamma = M^4 [\ln(\Lambda/\Lambda_{\mathrm{latt}})]^{-c},$$

where $\Lambda = a^{-1}$ is the ultraviolet cutoff, and Λ_{latt} and M are integration constants that are renormalization-group invariants. For the SU(n) group we have

$$b_0 = (4\pi)^{-2} \tfrac{1}{3}(11n - 2n_{\mathrm{f}}),$$

$$1 - c = \tfrac{9}{3} n (11n - 2n_{\mathrm{f}})^{-1},$$

where n_f is the number of quark flavors in the fundamental representation of SU(n). When expressed in terms of renormalized quantities, the horizon condition fixes the dimensionless ratio,

$$R \equiv \Lambda_{\text{latt}}/M.$$

We have

$$\alpha^{1/2}\frac{\partial W}{\partial \alpha^{1/2}} \approx M^2 \frac{\partial W}{\partial M^2} = \Lambda_{\text{latt}}^2 \frac{\partial W}{\partial \Lambda_{\text{latt}}^2}$$

to the accuracy of the calculation given below.)

Because α is the only dimensionful parameter in the continuum theory, to demonstrate that an area law holds for the Wilson loop W, it is sufficient to show that, when the enclosed area A is very large, the quantity

$$V \equiv \alpha^{1/2}\frac{\partial W}{\partial \alpha^{1/2}} \tag{11.2}$$

is given by

$$V = -c\alpha^{1/2} A W, \tag{11.3}$$

with a positive proportionality constant c. Moreover, in this case the string tension is given by

$$K = c\alpha^{1/2}. \tag{11.4}$$

For an SU(n) theory in four euclidean dimensions, we shall show that V is of the form

$$V = -c\alpha^{1/2} A W + \text{remainder}, \tag{11.5}$$

where the first term is obtained by evaluating the most simple and obvious non-perturbative contributions to V, which we shall call V_{simple}. This shows that the long range correlations we have found are capable of producing confinement, in the sense that the theory is confining if the dominant contributions to V are of the form of the simple non-perturbative contributions evaluated here. However, in the absence of a reliable estimate of the remainder, it cannot be said that confinement is established.

The parameter α appears explicitly only in the Boltzmann factor $\exp(-\alpha H)$, and we have

$$V_{\text{simple}} = -2\alpha\langle H \ \text{tr}\ U_p\rangle_c, \tag{11.6}$$

where the dependence on α contained in $g = g_{cr}(\alpha)$ is neglected, and the subscript c means connected. In order to simplify the presentation, we shall at this point convert to the continuum theory. The reader will have no difficulty in transcribing the following manipulations to the lattice theory. In the continuum, the covariant derivative is given by $D_\mu^{ac} = \partial_\mu \delta^{ac} + f^{abc}A_\mu^b$, the transporter by $U_p = P\exp(\int A_\lambda \, dx^\lambda)$, and the horizon function is

$$H = \int d^4x \, d^4y \, D_\nu^{ab}(x) D_\nu^{ac}(y) M^{-1\,bc}(x, y) - 4(n^2-1)V, \tag{11.7}$$

where V is the euclidean volume, and the constant term is the continuum limit of $(n^2-1)E(U)$ for $D=4$ euclidean dimensions. We now retain in V_{simple} the contribution that comes from the dipole singularity found in the preceding section. It occurs when $M^{-1\,bc}(x, y)$ is in disconnected graphs.

$$V_{\text{simple}} = -2\alpha \int d^4x \, d^4y \left\langle \mathrm{tr}\left[P\exp\left(\int A_\lambda \, dx^\lambda\right)\right] D_\nu^{ab}(x) D_\nu^{ac}(y) \right\rangle_c \times \langle M^{-1\,bc}(x, y)\rangle. \tag{11.8}$$

The term in $D_\nu(x) D_\nu(y)$ given by $\partial_\nu^{(x)} \partial_\nu^{(y)}$ is a completely disconnected part and does not contribute to V, the terms $\partial_\nu^{(x)} A_\nu(y)$ and $A_\nu(x)\partial_\nu^{(y)}$ give zero by global color symmetry, and we are left with

$$V_{\text{simple}} = -2n\alpha \int d^4x \, d^4y \left\langle A_\nu^a(x) A_\nu^a(y) \, \mathrm{tr}\left[P\exp\left(\int A_\lambda \, dx^\lambda\right)\right]\right\rangle_c \times \boldsymbol{G}(x-y), \tag{11.9}$$

where $\langle M^{-1\,bc}(x, y)\rangle = \delta^{bc}\boldsymbol{G}(x-y)$.

The contribution to V_{simple} of lowest connectivity, namely when the transporter is connected to a three-gluon vertex by a single gluon propagator, vanishes by global color invariance. We retain in V_{simple} the contribution to the expectation value which is of next lowest connectivity, namely when $A(x)$ and $A(y)$ are each connected to the transporter by a single gluon propagator,

$$\left\langle A_\nu^a(x) A_\nu^a(y) \, \mathrm{tr}\left[P\exp\left(\int A_\kappa \, dz^\kappa\right)\right]\right\rangle$$

$$\to \int du^\lambda \int dv^\mu \, \boldsymbol{D}_{\nu\lambda}(x-u) \boldsymbol{D}_{\nu\mu}(y-v) \left\langle \mathrm{tr}\left[t^1 U_1(u, v) t^a U_2(v, u)\right]\right\rangle, \tag{11.10}$$

where $\boldsymbol{D}_{\mu\nu}(x)\delta^{bc} = \langle A_\mu^b(x) A_\nu^c(0)\rangle$ is the gluon propagator, $\int du^\lambda$ and $\int dv^\mu$ are line integrals around the closed curve p, and $U_1(u, v)$ and $U_2(v, u)$ are the transporters corresponding to the two different pieces into which the closed curve $p = p_1 + p_2$ is decomposed,

$$U_1(u, v) = P\exp\left(\int_u^v A_\kappa \, dz^\kappa\right) \quad \text{on } p_1,$$

$$U_2(v, u) = P\exp\left(\int_v^u A_\kappa \, dz^\kappa\right) \quad \text{on } p_2. \tag{11.11}$$

In momentum space this reads

$$V_{\text{simple}} = -2n\alpha(2\pi)^{-4}\int d^4k\, D_{\lambda\nu}(k)G(k)D_{\nu\mu}(k)\int du^\lambda \int dv^\mu$$
$$\times \exp[ik\cdot(u-v)]\langle \operatorname{tr}[t^a U_1(u, v)t^a U_2(v, u)]\rangle. \tag{11.12}$$

From the identity

$$\sum_a t^a_{ij}t^a_{ml} = (2n)^{-1}\delta_{ij}\delta_{lm} - 2^{-1}\delta_{il}\delta_{jm}, \tag{11.13}$$

which holds for the basis matrices t^a of the fundamental representation of the Lie algebra of SU(n), we have

$$2n\langle \operatorname{tr}[t^a U_1(u, v)t^a U_2(v, u)]\rangle$$
$$= W - n\langle \operatorname{tr}[U_1(u, v)]\ \operatorname{tr}[U_2(v, u)]\rangle. \tag{11.14}$$

Here we have written

$$W = \langle \operatorname{tr}[U_1(u, v)U_2(v, u)]\rangle \tag{11.15}$$

for the gauge-invariant Wilson loop corresponding to the closed curve p. It is independent of u and v.

Which of the two terms in eq. (11.14) is larger, is difficult to say. For a $U(n)$ group, the first term is missing entirely. For an SU(n) theory and to zeroth order in the weak-coupling expansion, the second term is larger than the first by a factor n^2 and of opposite sign. However, expectation values of transporters along long curves are small quantities that result from delicate cancellations among low and very high order terms. In the absence of evidence to the contrary, the first term in eq. (11.14) may be dominant in an SU(n) theory, or may not be completely cancelled by the second term, and we shall retain it in V_{simple}.

(To illustrate this last point we exhibit simple examples in which the first term in eq. (11.14) is dominant. Let the probability distribution as a function of U_1 and

U_2, after all other variables are integrated out, be written $P(U_1, U_2)$, so the right hand side of eq. (11.14) is of the form

$$E \equiv \int dU_1\, dU_2\, P(U_1, U_2)\left[\mathrm{tr}(U_1 U_2) - n\,\mathrm{tr}\,U_1\,\mathrm{tr}\,U_2\right],$$

where dU represents the Haar measure for the SU(n) group. For the SU(2) group we use coordinates $U_1 = x^0 + i\boldsymbol{\sigma}\cdot\boldsymbol{x}$ and $U_2 = y^0 + i\boldsymbol{\sigma}\cdot\boldsymbol{y}$, so

$$\mathrm{tr}\,U_1 = 2x_0, \quad \mathrm{tr}\,U_2 = 2y_0, \quad \mathrm{tr}(U_1U_2) = 2\left(x^0y^0 - \boldsymbol{x}\cdot\boldsymbol{y}\right),$$

$$E = 2\left\langle P(x, y)\left[\left(x^0y^0 - \boldsymbol{x}\cdot\boldsymbol{y}\right) - 4x^0y^0\right]\right\rangle,$$

where the expectation value here means spherical averages over the two three-spheres $x^2 = y^2 = 1$. Global gauge invariance implies $P(x, y) = P(x^0, y^0, \boldsymbol{x}\cdot\boldsymbol{y})$. If P is of the form $P = f(\boldsymbol{x}\cdot\boldsymbol{y})$, then the second term in E vanishes. If P is of the form $P = \exp[\varepsilon(ax^0 + by^0 + c\boldsymbol{x}\cdot\boldsymbol{y})]$, where ε is a small parameter, then the first term in E is of order ε and the second term is of order ε^2. Etc.)

We retain in V_{simple} the first term in eq. (11.14) and obtain

$$V_{\text{simple}} = -\alpha W(2\pi)^{-4}\int du^\lambda \int dv^\mu \int d^4k$$
$$\times \exp\left[ik\cdot(u-v)\right]D^2(k)G(k). \tag{11.16}$$

Here we have written $D_{\mu\nu}(k) = D(k)P^{\mathrm{T}}_{\mu\nu}(k)$, where $P^{\mathrm{T}}_{\mu\nu}(k)$ is the transverse projector, and we have made the substitution

$$D_{\lambda\nu}(k)D_{\mu\nu}(k) - D^2(k)P^{\mathrm{T}}_{\lambda\mu}(k) \to D^2(k)\delta_{\lambda\mu}, \tag{11.17}$$

which is permissible inside the line integral around a closed curve, because the term proportional to $k_\mu k_\nu$ is a gradient.

As shown in the previous section, the Faddeev–Popov propagator $G(k)$ has the dipole singularity at $k = 0$, which in the continuum theory has the form

$$G(k) \sim c_1\alpha^{1/2}/(k^2)^2, \tag{11.18}$$

where $c_1 > 0$ is dimensionless. We now consider the case, discussed at the end of the previous section, that the gluon propagator at zero momentum $D(0)$ is a constant,

$$D(0) \sim c_2\alpha^{-1/2}, \tag{11.19}$$

where $c_2 > 0$ is dimensionless. (If $D(0)$ is infinite, then the theory would appear to exhibit a stronger confinement than by a string tension K. If $D(0) = 0$, we obtain the same expression given below for V_{simple}, by inserting identity (10.16) for M^{-1} into eqs. (11.6) and (11.7), keeping the dipole term, and then using the identity for the path ordered quantities,

$$\int \mathrm{d}s_2 \frac{\partial x^{\mu}(s_2)}{\partial s_2} ik_{\mu} \exp[ik\cdot x(s_2)]\left[t^a t^b \theta(s_1 - s_2) + t^b t^a \theta(s_1 - s_2)\right] \ldots$$

$$= \int \mathrm{d}s_2 \exp[ik\cdot x(s_2)](t^a t^b - t^b t^a)\delta(s_1 - s_2) \ldots + \ldots$$

$$= \exp[ik\cdot x(s_1)] f^{abc} t^c \ldots + \ldots, \tag{11.20}$$

to extract a local piece from higher order vertices.) Then for a large loop, the leading contribution to V_{simple} is given by

$$V_{\text{simple}} = -c_1(c_2)^2 \alpha^{1/2} W(2\pi)^{-4} \int \mathrm{d}u^{\mu} \int \mathrm{d}v^{\mu} \int \mathrm{d}^4 k$$
$$\times (k^2)^{-2} \exp[ik\cdot(u - v)]. \tag{11.21}$$

Let the closed curve lie in a plane, and choose a coordinate system adapted to this plane. The integral over the momentum components perpendicular to this plane gives

$$V_{\text{simple}} = -(16\pi^3)^{-1} c_1 c_2^2 \alpha^{1/2} W \int \mathrm{d}u^{\mu} \int \mathrm{d}v^{\mu} \int \mathrm{d}^2 k$$
$$\times (k^2)^{-1} \exp[ik\cdot(u - v)]. \tag{11.22}$$

Applying Stokes theorem to both line integrals, and with $\varepsilon^{\mu\lambda}\varepsilon^{\mu\nu} = \delta^{\lambda\nu}$, we obtain

$$V_{\text{simple}} = -(8\pi^2)^{-1} c_1 c_2^2 \alpha^{1/2} W \int \mathrm{d}^2 u \int \mathrm{d}^2 v\ \delta^2(u - v),$$

$$V_{\text{simple}} = -(8\pi^2)^{-1} c_1 c_2^2 \alpha^{1/2} A W, \tag{11.23}$$

where A is the area enclosed by the loop.

It is a pleasure to recall many stimulating discussions with Richard Brandt, Gianfausto Dell'Antonio, Stefano Fachin, Nicola Maggiore, Claudio Parrinello, Massimo Porrati, Jozef Namyslowski, Manfred Salmhofer, Martin Schaden, Erhard Seiler, Alan Sokal, and a correspondence with Pierre van Baal.

Appendix A

DISTANCE FUNCTION ON LATTICE CONFIGURATIONS AND UNIQUENESS OF ABSOLUTE MINIMUM

Let U and V be two lattice configurations, and consider the quantity $d(U, V)$ defined by

$$d^2(U, V) \equiv \sum_{\ell} \operatorname{tr}\left([U_\ell - V_\ell]^\dagger [U_\ell - V_\ell]\right), \tag{A.1}$$

where the sum extends over all links ℓ of the lattice, and the link variables are elements of SU(n). It obviously satisfies the defining properties of a distance function, in particular the triangle inequalities, for any set of matrices, not necessarily forming a group. This distance is gauge invariant, $d(U, V) = d(U^g, V^g)$.

If the configuration V is the vacuum configuration, $V_\ell = 1$ on each link ℓ, then we have

$$d^2(U, 1) = 2\sum_{\ell} (n - \operatorname{Re} \operatorname{tr} U_\ell) = 2nI(U), \tag{A.2}$$

where $I(U)$ is the minimizing function, eq. (1.3). Thus we may say that in the minimal lattice Landau gauge, which is the gauge used in this article, the representative configuration chosen on each gauge orbit is the one whose distance from the vacuum configuration is least.

This prescription fixes the gauge only to the extent that the minimum is unique. In this appendix we shall show that the interior of the set Λ of absolute minima consists of non-degenerate absolute minima. Our method will be to first relax the non-linear group properties, and to consider the minimization problem on the linear span of the group. This will allow us to completely solve the uniqueness problem using only the linear properties of the group matrices. We then impose the non-linear group properties as constraints on the absolute minima.

Let the structure group G be a unitary group, such as SU(n). The real linear span R(G) of the group G is the linear vector space which consists of matrices of the form $b = \sum_i a_i u_i$, where the a_i are real numbers, and $u_i \in$ G. As an example, the real linear span of the SU(2) group is the set of matrices of the form $b = b_0 - i\sum_{k=1}^{3} \sigma_k b_k$, where b_0 and b_k are real numbers. The SU(2) group is recovered by imposing the non-linear constraint $\sum_{a=0}^{3} b_a^2 = 1$, which defines an embedding of the unit sphere S_3 in $\mathbb{R}_4$. We define an extended lattice configuration B by assigning an element $b_{xy} \in$ R(G) to each link $\ell = (xy)$ of the lattice, where x and y are nearest neighbors, and we write $B = \{b_{xy}\}$. In lattice gauge theory, the condition $u_{yx} = u_{xy}^{-1} = u_{xy}^\dagger$ holds for unitary groups, so for the real linear span, the b_{xy} satisfy $b_{yx} = b_{xy}^{\dagger}$. The set of extended configurations form a real vector space V in the obvious way,

$$a_1 B_1 + a_2 B_2 = \{a_1 b_{1xy} + a_2 b_{2xy}\}.$$

The local gauge transform B^g of an extended configuration $B = \{b_{xy}\}$ by the local gauge transformation $g = \{g_x\}$ is the configuration $B^g \equiv \{g_x^{\dagger} b_{xy} g_y\}$, where $g_x \in$ G. (The gauge transformations g_x remain in the group, and are *not* extended to the linear span.) The original, unextended configurations, for which the non-linear constraints hold, form a smooth non-linear submanifold M of the vector space V that is compact and without a boundary. Note incidentally that in order to define an effective action in lattice gauge theory, we have found it necessary to write the partition function as an integral over the linear span, with the non-linear conditions imposed as constraints that are written explicitly in sect. 9 for the SU(2) group, and in Appendix G for the SU(n) group.

Consider the homogeneous linear function on extended configurations defined by

$$K(B) \equiv n^{-1} \sum_{(xy)} \mathrm{Re}\ \mathrm{tr}\ b_{xy}, \tag{A.3}$$

which extends the maximizing function, eq. (1.2b). We take as a maximizing function, the restriction of this function to the gauge orbit through an arbitrary fixed configuration B,

$$F_B(g) \equiv K(B^g). \tag{A.4}$$

For a finite lattice, the gauge orbit is compact because the g_x are unitary matrices, and for fixed B, the maximizing function $F_B(g)$ is bounded above. Consequently at least one maximizing gauge transformation h does exist on each gauge orbit. After gauge transformation by h, the maximum is brought to $g = 1$.

We define the extended fundamental modular region Λ_{E} to be the set of extended configurations that are absolute maxima on each gauge orbit,

$$\Lambda_{\mathrm{E}} \equiv \{B\colon F_B(1) \geqslant F_B(g)\}. \tag{A.5}$$

For constant gauge transformations, namely $g_x = g_y$ for all x and y, we have $F_B(g) = F_B(1)$, so maxima are at most unique modulo constant gauge transformations. It will be seen shortly that the interior points Λ_{E} are unique absolute maxima, modulo constant gauge transformations. The original fundamental modular region Λ, is regained from the intersection

$$\Lambda = \Lambda_{\mathrm{E}} \cap \mathrm{M}, \tag{A.6}$$

where M is the submanifold of the vector space V where the non-linear group constraints hold.

The proofs of the following theorems are given below.

Theorem A.1. The set Λ_E forms a convex cone, with vertex given by the configuration $B = 0$. Its interior points are unique absolute maxima of the maximizing function (A.4), modulo constant gauge transformations.

Theorem A.2. The interior of the set Λ consists of absolute maxima that are unique modulo constant gauge transformations.

Remark 1. At an absolute maximum, the maximizing function (A.4) is stationary. It is easy to show, by a straightforward extension of the proof given in Appendix B, that if $F_B(g)$ is stationary at $g = 1$, then B satisfies the lattice transversality condition

$$\sum_{y \in (xy)} \mathrm{tr}\left[t^a\left(b_{xy} - b_{xy}^\dagger\right)\right] = 0. \tag{A.7}$$

Here x is fixed but arbitrary, the sum is over the nearest neighbors y or x, and the t^a form a basis of the Lie algebra of G. This condition is linear in B and thus defines a hyperplane Γ_E in the vector space of extended configurations V, that contains Λ_E.

$$\Lambda_E \subset \Gamma_E. \tag{A.8}$$

Remark 2. On the boundary $\partial\Lambda_E$, the maxima are not unique in general. The degenerate maxima on $\partial\Lambda_E$ must be identified, and it is only after this identification that Λ_E becomes the extended orbit space. To this extent the gauge-fixing is non-unique.

Proof of theorem A.1. We first show that Λ_E is a convex cone. Let α and β be positive numbers, and let B_1 and B_2 be in Λ_E. We have

$$\begin{aligned} K(\alpha B_1 + \beta B_2) &= \alpha K(B_1) + \beta K(B_2) \\ &\geqslant \alpha K(B_1^g) + \beta K(B_2^g) = K\left[(\alpha B_1 + \beta B_2)^g\right], \end{aligned}$$

where g may be any local gauge transformation. Thus $\alpha B_1 + \beta B_2$ is in Λ_E. By choosing $\beta = 0$, it follows that Λ_E is a cone with the configuration $B = 0$ at the vertex, and by choosing $\alpha + \beta = 1$, it follows that Λ_E is convex.

We now prove that all interior points of Λ_E are unique absolute maxima, following the method of proof given in the continuum case by van Baal [4]. Let

$U_0 \in \mathrm{V}$ be the "vacuum configuration" defined by $U_0 = \{v_{xy}\}$, with $v_{xy} = 1$ for all links (xy). The vacuum configuration U_0 lies in Λ_{E}, for we have

$$K(U_0^g) = \sum_{(xy)} \mathrm{Re}\ \mathrm{tr}\left(g_x^\dagger g_y\right) \leqslant \sum_{(xy)} \mathrm{Re}\ \mathrm{tr}\ 1 = K(U_0).$$

Note that the inequality is strict,

$$K(U_0^g) < K(U_0), \tag{A.9}$$

unless $g_x = g_y$ for all pairs of nearest neighbors x and y, in which case $U_0^g = U_0$. Consequently, the vacuum configuration U_0 is a unique absolute maximum. Let B be any other configuration that lies in the interior of Λ_{E}. Because Λ_{E} is convex, the line segment joining U_0 and B – which consists of the set of points $\alpha U_0 + \beta B$ with α and β positive and $\alpha + \beta = 1$ – lies in Λ_{E}. Because B is an interior point of Λ_{E}, this line segment may be extended beyond B to some other point B_1 that also lies in Λ_{E}. Therefore we may write $B = \alpha U_0 + \beta B_1$, for some positive α and β, with $\alpha + \beta = 1$, so we have

$$\begin{aligned} K(B^g) &= \alpha K(U_0^g) + \beta K(B_1^g) \\ &< \alpha K(U_0) + \beta K(B_1) = K(\alpha U_0 + \beta B_1) = K(B). \end{aligned}$$

The inequality is strict for non-constant g_x by eq. (A.9), so B is a unique maximum. QED.

Proof of theorem A.2. Whereas Λ_{E} is a convex cone with a boundary $\partial\Lambda_{\mathrm{E}}$, the non-linear manifold M, which consists of the configurations whose link variables are group elements, is a smooth submanifold of V without a boundary. Therefore the boundary $\partial\Lambda$ and the interior Λ^0 of the intersection $\Lambda = \Lambda_{\mathrm{E}} \cap \mathrm{M}$ are given by the corresponding intersections

$$\begin{aligned} \partial\Lambda &= \partial\Lambda_{\mathrm{E}} \cap \mathrm{M}, \\ \Lambda^0 &= \Lambda_{\mathrm{E}}^0 \cap \mathrm{M}, \end{aligned} \tag{A.10}$$

where Λ_{E}^0 is the interior of Λ_{E}, unless the intersection is singular in the sense that either (a) the intersection $\Lambda = \Lambda_{\mathrm{E}} \cap \mathrm{M}$ is entirely contained in the boundary $\partial\Lambda_{\mathrm{E}}$, or (b) some proper subset of $\Lambda = \Lambda_{\mathrm{E}} \cap \mathrm{M}$, that is of the same dimension as Λ itself, is contained in $\partial\Lambda_{\mathrm{E}}$. We conclude that the interior points of the fundamental modular region Λ are unique absolute maxima of $E(U)$, as asserted, provided that the singular cases (a) or (b) do not hold.

We now show that case (a) does not hold. To prove this, observe that the vacuum configuration, $U_0 = \{v_{xy}\}$ with $v_{xy} = 1$, is contained in Λ, so it is sufficient to prove that U_0 is an interior point of Λ_E. This will be true if, for a sufficiently small neighborhood of V in Γ_E, namely for all configurations of the form $B_1 = U_0 + B$, with B in Γ_E and $|B|^2 < \delta$, we have $K(B_1^g) < K(B_1)$, for sufficiently small δ and for all local gauge transformations g, where δ is uniform in g. This is not hard to prove using eq. (A.7). This proves incidentally that Γ_E and Λ_E have the same dimension.

To complete the proof, we now show that case (b) does not hold either. Let Γ be the intersection

$$\Gamma = \Gamma_E \cap \mathrm{M}. \tag{A.11}$$

Because Λ_E is contained in Γ_E, we have

$$\Lambda = \Lambda_E \cap \Gamma. \tag{A.12}$$

If case (b) holds then Γ must merge into the boundary $\partial\Lambda_E$. However, Γ is a smooth manifold that has a neighborhood that is contained in Λ_E, as we have just shown. On the other hand Λ_E is a convex cone. Therefore Γ cannot merge with the boundary $\partial\Lambda_E$ unless it has a kink where it meets the boundary. But Γ is the intersection of a hyperplane Γ_E with M which is a product of group manifolds. Therefore it has no kink. QED.

Appendix B

LATTICE FADDEEV–POPOV OPERATOR

In this appendix we shall develop formulas for the derivatives of $F_U(g)$ along one-parameter subgroups of the local gauge group, for the lattice gauge-covariant derivative, the lattice Faddeev–Popov operator and for the hessian of $F_U(g)$.

Consider the one-parameter subgroup $g(\tau)$ of the local gauge group defined by

$$g(\tau) = \{g(\tau, x)\} = \{\exp[\tau\omega(x)]\}, \quad \omega(x) = -\omega^\dagger(x), \tag{B.1}$$

where $\omega(x)$ is a generic element of the local Lie algebra, $\omega(x) = t^a\omega^a(x)$, and the t^a form the anti-hermitian basis of the Lie algebra which was defined in the introduction. For fixed ω and U, let $U(\tau)$ be the one-parameter curve on the gauge orbit through U defined by

$$U_\mu(\tau, x) \equiv g^\dagger(\tau, x)U_\mu(x)g(\tau, x + e_\mu), \tag{B.2}$$

where e_μ is a unit vector in the positive μ-direction, and let $F(\tau)$ be defined by

$$F(\tau) \equiv F_U[g(\tau)] = I[U(\tau)], \tag{B.3}$$

where $F_U(g)$ and $I(U)$ are defined in eqs. (1.3) and (1.2a). We have

$$F'(\tau) = -n^{-1} \sum_{x,\mu} \operatorname{Re} \operatorname{tr}\left\{\left[\omega(x+e_\mu) - \omega(x)\right] U_\mu(\tau, x)\right\},$$

$$F'(\tau) = -(2n)^{-1} \sum_x \omega^a(x) \sum_\mu \left[A_\mu^a(\tau, x) - A_\mu^a(\tau, x-e_\mu)\right], \tag{B.4}$$

where $A_\mu^a(x)$ is defined in eq. (1.6). We write this as

$$F'(\tau) = -(2n)^{-1}(\omega, \nabla \cdot A(\tau)), \tag{B.5}$$

where $\nabla \cdot A$ is the lattice divergence of A,

$$\nabla \cdot A^a(x) \equiv \sum_\mu \left[A_\mu^a(x) - A_\mu^a(x-e_\mu)\right]. \tag{B.6}$$

If U is a stationary point of the minimizing function at $g=1$, we have, $F'(0)=0$ for all $\omega^a(x)$, which implies $\nabla \cdot A = 0$. The transversality of $A(U)$ is thereby established as the condition for the minimizing function $F_U(g)$ to be stationary at $g=1$. This equation defines the hyperplane Γ in A-space,

$$\Gamma \equiv \{U: \nabla \cdot A(U) = 0\}. \tag{B.7}$$

The second derivative of the minimizing function is given by

$$F''(\tau) = -(2n)^{-1}(\omega, \nabla \cdot A'(\tau)). \tag{B.8}$$

Here $A'(\tau)$ is linear in ω,

$$A_\mu'^a(\tau, x) = \left[D_\mu(U(\tau))\omega\right]^a(x). \tag{B.9}$$

This expression defines $D_\mu(U)\omega$, which is the lattice gauge-covariant derivative of ω in the adjoint representation. (It is the Lie derivative of U in the ω-direction, $L_\omega U$.) Geometrically, it is the vector which is tangent to the curve $U(\tau)$, eq. (B.2), in the A coordinate basis. Its components may be found as

$$\begin{aligned} A_\mu'^a(\tau, x) &= -2 \operatorname{Re} \operatorname{tr}\left[t^a U_\mu'(\tau, x)\right] \\ &= -2 \operatorname{Re} \operatorname{tr}\left\{\left[\omega(x+e_\mu)t^a - t^a\omega(x)\right] U(\tau, x)\right\} \\ &= -\tfrac{1}{2} \operatorname{tr}\Big(\left\{\left[\omega(x+e_\mu) - \omega(x)\right], t^a\right\}\left[U(\tau, x) + U^\dagger(\tau, x)\right] \\ &\quad + \left[\left[\omega(x+e_\mu) + \omega(x)\right], t^a\right]\left[U(\tau, x) - U^\dagger(\tau, x)\right]\Big). \end{aligned} \tag{B.10}$$

This gives the explicit form of the gauge-covariant derivative,

$$[D_\mu(U)\omega]^a(x) = G_\mu^{ab}(x)[\omega(x+e_\mu) - \omega(x)]^b + \tfrac{1}{2}f^{abc}A_\mu^b(x)[\omega(x+e_\mu) + \omega(x)]^c, \tag{B.11}$$

where the real symmetric matrix $G_\mu^{ab}(x)$ is defined by

$$G_\mu^{ab}(x) \equiv -\tfrac{1}{2}\operatorname{tr}\{(t^a t^b + t^b t^a)[U_\mu(x) + U_\mu^\dagger(x)]\}, \tag{B.12a}$$

$$G_\mu^{ab}(x) = \tfrac{1}{2}\operatorname{tr}\{(n^{-1}\delta^{ab} + i d^{abc}t^c)[U_\mu(x) + U_\mu^\dagger(x)]\}. \tag{B.12b}$$

From $\sum_a (t^a)^2 = -(n^2-1)(2n)^{-1}$ we have also

$$\sum_a G_\mu^{aa}(x) = (n^2-1)n^{-1}\operatorname{Re}\operatorname{tr}[U_\mu(x)]. \tag{B.12c}$$

For the special case $U_\mu(x) = 1$, we have $G_\mu^{ab}(x) = \delta^{ab}$, and for the SU(2) group we have

$$G_\mu^{ab}(x) = \pm\left[1 - \left|\tfrac{1}{2}A_\mu(x)\right|^2\right]^{1/2}\delta^{ab}. \tag{B.13}$$

The upper sign holds in the U_+ coordinate patch.

The second derivative of $F(\tau)$,

$$F''(\tau) = -(2n)^{-1}(\omega, \nabla\cdot(D(U)\omega)), \tag{B.14}$$

defines a symmetric quadratic form,

$$(\omega, M(U)\omega) \equiv -(\omega, \nabla\cdot(D(U)\omega)), \tag{B.15}$$

whose kernel $M(U)$ is the hessian of $F_U(g)$. It is a real symmetric matrix given by the symmetric part of the lattice Faddeev–Popov operator, $-\nabla\cdot D(U)$,

$$M(U)\omega = -2^{-1}[\nabla\cdot(D(U)\omega) + D(U)\cdot(\nabla\omega)]. \tag{B.16}$$

The Faddeev–Popov operator is symmetric when $A(U)$ is divergenceless,

$$\nabla\cdot A = 0 \leftrightarrow M(U) = -\nabla\cdot D(U) = -D(U)\cdot\nabla. \tag{B.17}$$

In this case, the hessian $M(U)$ acts according to

$$(M\omega)^a(x) = \sum_\mu \Big\{G_\mu^{ab}(x)[\omega^b(x) - \omega^b(x+e_\mu)] - (x \leftrightarrow x - e_\mu) - \tfrac{1}{2}f^{abc}[A_\mu^b(x)\omega^c(x+e_\mu) - A_\mu^b(x-e_\mu)\omega^c(x-e_\mu)]\Big\}, \tag{B.18}$$

where $G_\mu^{ab}(x)$ is given in eq. (B.12).

From eqs. (B.2) and (B.3), one has

$$F'''(\tau) = -n^{-1} \sum_{x,\mu} \mathrm{Re}\ \mathrm{tr}\Big\{\big[\omega^3(x+e_\mu) - 3\omega^2(x+e_\mu)\omega(x)$$

$$+3\omega(x+e_\mu)\omega^2(x) - \omega^3(x)\big]U_\mu(\tau,x)\Big\}, \tag{B.19}$$

for the SU(n) group. For the SU(2) group this may be written

$$F''' = \tfrac{3}{16} \sum_{x,\mu} \big[\omega^2(x+e_\mu) - \omega^2(x)\big]\big[\omega(x+e_\mu) + \omega(x)\big] \cdot A_\mu(\tau,x)$$

$$+ \tfrac{1}{4} \sum_x \omega^2(x)\omega(x) \cdot \sum_\mu \big[A_\mu(\tau,x) - A_\mu(\tau,x-e_\mu)\big], \tag{B.20}$$

where the dot product means contraction in color space and $\omega^2(x) \equiv \omega(x)\cdot\omega(x)$. From eq. (B.11) for the gauge-covariant derivative we obtain the fourth derivative, in the case of the SU(2) group

$$F'''' = \tfrac{3}{16} \sum_{x,\mu} \big[\omega^2(x+e_\mu) - \omega^2(x)\big]^2 G_\mu(x)$$

$$+ \tfrac{1}{4} \sum_x \omega^2(x)\omega(x) \cdot \big[\nabla\cdot(D(U)\omega)\big]. \tag{B.21}$$

The geometric interpretation of these quantities when $U = U(A)$ lies on $\partial\Omega$ and ω is in the null-space of the hessian, $M(U)\omega = 0$, may be seen as follows. At any point on the hyperplane Γ, the first derivative $(\nabla\cdot A, \omega)$ of the minimizing function $F_U(g)$ vanishes in all gauge orbit directions ω, so F' vanishes on $\partial\Omega \subset \Gamma$. Let $\delta A = \varepsilon(D(U)\omega)$, where ε is a small quantity. The gauge orbit in the ω-direction is tangent to Γ provided that the condition

$$\nabla\cdot\delta A = \varepsilon\nabla\cdot(D(U)\omega) = -\varepsilon M(U)\omega = 0 \tag{B.22}$$

holds. This shows that at every point U on $\partial\Omega$, the gauge orbit is tangent to Γ. This is equivalent to the vanishing of the second derivative $F'' = (\omega, M(U)\omega)$. The boundary $\partial\Omega$ is determined by the equation $\lambda(U) = 0$, where $\lambda(U)$ is the lowest eigenvalue of the hessian $M(U)$. Let $\delta\lambda = \lambda(A+\delta A) - \lambda(A)$ for $\delta A = \varepsilon(D(A)\omega)$. The gauge orbit is also tangent to $\partial\Omega$ provided the condition

$$\delta\lambda = (\partial\lambda/\partial A_i, \delta A_i) = 0 \tag{B.23}$$

also holds, where $\{A_i\} = \{A_\mu^a(x)\}$. The expression

$$\lambda(A) = (\omega, M(A)\omega)/(\omega, \omega)$$

is stationary with respect to first-order variations of ω, so that

$$\delta\lambda(A) = (\omega, \delta M(A)\omega)/(\omega, \omega)$$

(even though $\omega = \omega(A)$ depends on A). With $F(\tau) = F_U[g(\tau)]$ where $g(\tau) = \exp(\omega\tau)$, we have

$$\delta\lambda = \varepsilon F'''/(\omega, \omega), \tag{B.24}$$

so the vanishing of F''' at a point on $\partial\Omega$ means that the gauge orbit in the ω direction is tangent to $\partial\Omega \subset \Gamma$.

Appendix C

CONFIGURATION ON DISTINGUISHED BOUNDARY

Consider an SU(n) gauge theory on a periodic lattice in D dimensions, with edges $L_1, L_2, \ldots, L_D$. Every SU(n) group contains an SU(2) subgroup. Let U be the configuration in the SU(2) subgroup given by

$$\begin{aligned} U_1(x) &= \cos\beta + 2t^3(-1)^{x_2}\sin\beta, \\ U_2(x) &= \cos\beta + 2t^3(-1)^{x_1}\sin\beta, \\ U_\mu(x) &= 1 \quad \text{for } \mu > 2. \end{aligned} \tag{C.1}$$

It is transverse, $\nabla \cdot A(U) = 0$. For this configuration to be periodic, the edges L_1 and L_2 must be even, and the other edges are arbitrary integers. Without loss of generality, we suppress dimensions $D > 2$.

We shall now prove that configuration (C.1) lies in the core of the fundamental modular region Ξ_L for $\beta < |\frac{1}{4}\pi|$. The Wilson loop about a single plaquette in the 1–2 plane, $W_p = \frac{1}{2}$ tr U_p, has either of the two values

$$W_p = \cos(4\beta) \quad \text{or} \quad W_p = 1, \tag{C.2}$$

and these values alternate in checkerboard fashion on the red and black plaquettes. Consider the new problem of finding the minimum of the minimizing function $I(U) = \Sigma_\ell(1 - \frac{1}{2}\text{ tr } U_\ell)$, on a lattice edge NL_1, NL_2, subject only to the constraints that $W_p = \cos(4\beta)$ on red plaquettes. Each link of the lattice borders one and only one red plaquette with the value $W_{\mathrm{p}} = \cos(4\beta)$, so this minimizing problem decomposes into minimizing independently on each red plaquette. A moment's reflection shows that, for β in the interval

$$-\tfrac{1}{4}\pi \leqslant \beta \leqslant \tfrac{1}{4}\pi, \tag{C.3}$$

the absolute minimum of the new minimizing problem is given by the original configuration (C.1). It follows that, for this range of β, configuration (C.1) also gives the absolute minimum of the minimizing function on its gauge orbit, because the minimizing function cannot possibly have a lower value, when the values (for this configuration) of the Wilson loops that are associated with all other closed paths are imposed as additional constraints. This is true for all N, and consequently configuration (C.1) lies in the core of the fundamental modular region of the original lattice Ξ_L, as stated.

We now show that at $\beta = \pm\frac{1}{4}\pi$, the configuration (C.1) saturates the bound (3.14), $0 \leqslant H(U, \eta) \leqslant 0$, where $H(U, \eta)$ is given in eq. (3.11). We require $B^a_{\mu c}(x)$ which is defined in eq. (3.7b). For the configuration (C.1), we have

$$B^a_{1c}(x) = \varepsilon^{a3c} 2(-1)^{x_2} \sin\beta,$$

$$B^a_{2c}(x) = \varepsilon^{a3c} 2(-1)^{x_1} \sin\beta. \tag{C.4}$$

From eq. (B.18) for the lattice Faddeev–Popov operator $M(U)$, we find

$$[M(U)B]^a_{\mu c}(x) = 4\cos\beta\ B^a_{\mu c}(x),$$

and hence

$$\left[M^{-1}(U)B\right]^a_{\mu c}(x) = (4\cos\beta)^{-1} B^a_{\mu c}(x). \tag{C.5}$$

Thus the horizon tensor, defined in eq. (3.12b), is given by

$$H^{cd}_{\mu,\nu}(U) = V\delta_{\mu\nu}\delta^{cd}\left[-\cos\beta + (1-\delta^{c3})(\cos\beta)^{-1}\sin^2\beta\right]. \tag{C.6}$$

With $\eta^a_\mu = n_\mu v^a$, and for any unit euclidean vector n in the 1–2 plane, and any unit color vector v in the 1–2 color plane, we have

$$H(U, \eta) = V\left[-\cos\beta + (\cos\beta)^{-1}\sin^2\beta\right]. \tag{C.7}$$

at $|\beta| = \frac{1}{4}\pi$, the bound, $H(U, \eta) \leqslant 0$, is saturated and the configuration lies on the distinguished boundary, as asserted. QED.

Appendix D

BRILLOUIN ZONE OF GRIBOV COPIES FOR CONFIGURATIONS OF ALL FREQUENCIES

Consider the SU(2) configuration

$$U_\mu(x) = 1 \quad \text{for } \mu \neq 2,$$

$$U_2(x) = \cos\beta + 2t \cdot u(x^1)\sin\beta,$$

$$u(x^1) \equiv \cos(2\pi m x^1/L)e_1 + \sin(2\pi m x^1/L)e_2, \tag{D.1}$$

where e_i are an orthonormal triplet of color vectors. In order to determine whether this configuration lies inside the core region Ξ_L, we shall compare $I(U)$ with $I(U^g)$, where the minimizing gauge transformation $g(x)$ is of the form given in theorem 1, eq. (2.7). We ignore directions $\mu > 2$. This configuration is invariant under all translations in the 2-direction, so according to sect. 2, theorem 1 and remark 3, $g(x)$ is of the form

$$g(x) = h(x^1)\left[\cos(\theta_\mu x^\mu) + 2t \cdot v \sin(\theta_\mu x^\mu)\right], \tag{D.2}$$

$$\theta_\mu = 2\pi n_\mu/NL, \tag{D.3}$$

where $\mu = 1$ and 2. To determine the minimizing configuration $g(x)$, we make the ansatz $v = e_3$, where e_3 is a unit color vector in the 3-direction, and

$$h(x^1) = \cos\gamma + 2t \cdot w(x^1)\sin\gamma. \tag{D.4}$$

It will be found shortly that the unit color vector $w(x^1)$ is given by

$$w(x^1) = e^3 \times u(x^1) = -\sin(2\pi m x^1/L)e_1 + \cos(2\pi m x^1/L)e_2, \tag{D.5}$$

where $(p \times q)^a = \varepsilon^{abc}p^b q^c$, so e_3, $u(x^1)$ and $w(x^1)$ are a triplet of orthonormal color vectors. To determine γ and $w^a(x^1)$, we substitute these expressions into the transversality condition (2.10) and obtain, after conjugation with $h(x^1)$,

$$\begin{aligned} &h(x^1+1)\exp(\theta_1 2t^3)h^\dagger(x^1) + h(x^1-1)\exp(-\theta_1 2t^3)h^\dagger(x^1) \\ &+ U_2(x^1)h(x^1)\exp(\theta_2 2t^3)h^\dagger(x^1) - \exp(\theta_2 2t^3)h^\dagger(x^1)U_2(x^1) \\ &- \text{h.c.} = 0. \end{aligned} \tag{D.6}$$

This gives

$$\begin{aligned}&\left[h(x^1+1)+h(x^1-1)\right]h^\dagger(x^1)\cos\theta_1\\&\quad+\left[h(x^1+1)-h(x^1-1)\right]2t^3h^\dagger(x^1)\sin\theta_1\\&\quad+\sin\beta\sin\theta_2\left[2t\cdot u(x^1),2t\cdot p(x^1)\right]-\text{h.c.}=0,\end{aligned}\tag{D.7}$$

where

$$\begin{aligned}2t\cdot p(x^1)&\equiv h(x^1)2t^3h^\dagger(x^1)\\&=2\left\{\cos(2\gamma)t^3+\sin(2\gamma)2t\cdot\left[w(x^1)\times e^3\right]\right\}.\end{aligned}$$

Observe that eq. (D.7) is solved by $w(x^1)$ given in eq. (D.5), for we have

$$\begin{aligned}w(x^1+1)+w(x^1-1)&=2\cos(2\pi m/L)w(x^1),\\w(x^1+1)-w(x^1-1)&=-2\sin(2\pi m/L)u(x^1),\\2t\cdot\left[w(x^1+1)-w(x^1-1)\right]2t^3&=-2\sin(2\pi m/L)t\cdot u(x^1)2t^3\\&=2\sin(2\pi m/L)t\cdot w(x^1),\\\left[t\cdot u(x^1),t\cdot p(x^1)\right]&=t\cdot\left[u(x^1)\times p(x^1)\right],\\u(x^1)\times p(x^1)&=\cos(2\gamma)u(x^1)\times e^3\\&=-\cos(2\gamma)w(x^1).\end{aligned}$$

Thus all terms in eq. (D.7) are proportional to $w(x^1)$. Upon collecting terms, one obtains the angle γ,

$$\tan(2\gamma)=-2\left[\cos\theta_1-\cos(\alpha-\theta_2)\right]^{-1}\sin\beta\sin\theta_2,\tag{D.8}$$

where $\alpha\equiv 2\pi m/L$.

The form of the minimizing gauge transformation is now completely determined. From the condition $I(U)\leqslant I(U^g)$ for all configurations U in Ξ_L, one obtains the bound (2.26) by taking $\theta_1=0$ and θ_2 small.

Appendix E

CONFIGURATIONS ON A DEGENERATE GAUGE ORBIT

Consider an SU(n) gauge theory on a periodic lattice in D dimensions, with edges $L_1, L_2, \dots, L_D$. Every SU(n) group contains an SU(2) subgroup. Let U be the configuration in the SU(2) subgroup given by

$$U_1(x) = \cos\beta + 2t_3(-1)^{x_1+x_2}\sin\beta,$$

$$U_2(x) = \cos\beta - 2t_3(-1)^{x_1+x_2}\sin\beta,$$

$$U_\mu(x) = 1 \quad \text{for } \mu > 2. \tag{E.1}$$

Without loss of generality we suppress dimensions $\mu > 2$. The transporter about a single plaquette is given by

$$U_p(x) = U_1(x)U_2(x+e_1)U_1^\dagger(x+e_2)U_2^\dagger(x),$$

$$U_p(x) = \cos(4\beta) + 2t_3(-1)^{x_1+x_2}\sin(4\beta), \tag{E.2}$$

and we have

$$\tfrac{1}{2}\,\mathrm{tr}\big[U_p(x)\big] = \cos(4\beta). \tag{E.3}$$

By an argument similar to the one at the beginning of the preceding appendix, we conclude that configuration (E.1) lies in the core Ξ_L of the fundamental modular region for β in the range

$$-\tfrac{1}{4}\pi \leqslant \beta \leqslant \tfrac{1}{4}\pi. \tag{E.4}$$

Configuration (E.1) in two dimensions lies on a degenerate gauge orbit for $\beta = \frac{1}{4}\pi$, because the equation $D_\mu^{ac}(U)\omega^c](x) = 0$ possesses the solution $\omega^a(x) = e_\pm^a\varphi_\pm(x)$, where $e_\pm$ satisfies $e_3 \times e_\pm = \pm ie_\pm$, and

$$\varphi_\pm(x) = (1 \pm i)(-1)^x + [1 - (\pm i)](-1)^y, \tag{E.5}$$

with $x = x_1$ and $y = x_2$. Consequently ω is also a null eigenvector, $M\omega = 0$, of the Faddeev–Popov operator $M = -\nabla\cdot D(U)$, and thus lies on the boundary $\partial\Omega_L$. Therefore, since $\Xi_L \subset \Omega_L$, configuration (E.1) lies on the boundaries $\partial\Xi_L$ and $\partial\Omega_L$ for $\beta = \pm\frac{1}{4}\pi$, as asserted.

It would at first appear that the configuration (E.1) is an example of a boundary point that does not lie on the distinguished boundary. Observe that $B_{\mu c}^a(x)$, which is defined in eq. (3.7), vanishes identically for configuration (E.1). (In fact, in two dimensions this is the only configuration (modulo constant gauge transformations) with this property.) Consequently it would appear that the quadratic form (3.12) is given by

$$H_{\mu a,\nu b}(U) = -\delta_{\mu\nu}\sum_x G_\mu^{ab}(x) = -\delta_{\mu\nu}\delta^{ab}V\cos\beta,$$

which is strictly negative at $\beta = \pm \frac{1}{4}\pi$, whereas on the distinguished boundary $H_{\mu a,\nu b}$ has a vanishing eigenvalue. However, we have just seen that $M(U)$ has a zero eigenvalue at $\beta = \pm \frac{1}{4}\pi$, and consequently the first term in eq. (3.12) is really of the form 0/0, and its value depends on how the configuration (E.1) is approached. An investigation of an infinitesimal neighborhood of configuration (E.1) shows that it is approached by the distinguished boundary and thus does lie on it.

Appendix F

PROOF OF THEOREM 1

If $g(x)$ is a minimizing gauge transformation, then so is the translated gauge transformation ${}^{L\mu}g(x) \equiv g(x + Le_\mu)$, because the minimizing function $F_U(g)$ is invariant under translation by L when U is, $F_U({}^{L\mu}g) = F_U(g)$. On the other hand, as proven in Appendix A, absolute minima are unique, modulo x-independent gauge transformations, so we have

$$ {}^{L\mu}g(x) = g(x + Le_\mu) = g(x)k_\mu, \tag{F.1} $$

where, for each μ, k_μ is an x-independent element of the SU(n) group. Because translations commute, we have

$$ g(x + Le_\mu + Le_\nu) = g(x)k_\mu k_\nu = g(x)k_\nu k_\mu, $$

and thus the k_μ commute,

$$ k_\mu k_\nu = k_\nu k_\mu. $$

Moreover, $g(x)$ has period NL, and so

$$ g(x + NLe_\mu) = g(x)(k_\mu)^N = g(x), $$

which gives

$$ (k_\mu)^N = 1. $$

The most general solution to these equations for k_μ is given by

$$ k_\mu = \exp(t^a \varphi_\mu^a), $$

where the φ_μ^a lie in a Cartan subalgebra

$$ \left[t^b\varphi_\mu^b,\, t^c\varphi_\nu^c\right] = 0, $$

and for each μ have eigenvalues restricted by

$$t^a \varphi_\mu^a \psi^i = \left(2\pi i n_\mu^i / N\right) \psi^i.$$

The general form of $g(x)$ is most simply expressed by introducing coordinates y^μ and z^μ according to

$$x^\mu = L y^\mu + z^\mu, \quad 0 \leqslant y^\mu \leqslant N-1, 0 \leqslant z^\mu \leqslant L-1,$$

so y^μ labels blocks and z^μ labels sites within a block. Iteration of eq. (F.1) gives

$$g(x) = g_1(z)(k_\mu)^{y^\mu} = g_1(z) \exp\left(t^a \varphi_\mu^a y^\mu\right).$$

We define $h(z)$ by

$$g_1(z) = h(z) \exp\left(t^a \varphi_\mu^a z^\mu / L\right),$$

which gives

$$g(x) = h(z) \exp\left(t^a \varphi_\mu^a x^\mu / L\right),$$

and the theorem is established, with $\theta_\mu^a \equiv \varphi_\mu^a / L$.

According to eqs. (1.6) and (B.6) the minimizing configuration U^g is transverse, namely,

$$\tfrac{1}{2} \sum_\mu \left[g^\dagger(x) U_\mu(x) g(x+e_\mu) - g^\dagger(x+e_\mu) U_\mu^\dagger(x) g(x) \right.$$
$$\left. - g^\dagger(x-e_\mu) U_\mu(x-e_\mu) g(x) + g^\dagger(x) U_\mu^\dagger(x-e_\mu) g(x-e_\mu) \right]_{\text{traceless}} = 0.$$

We substitute the form (2.7) for $g(x)$ and multiply from the left by $\exp(t^a \theta_\mu^a x^\mu)$, and from the right by $\exp(-t^a \theta_\mu^a x^\mu)$. Using the fact that the trace is invariant under conjugation, eq. (2.10) follows. QED.

Appendix G

DECOMPACTIFICATION AND SOURCES FOR THE SU(n) GROUP

Let U be an element of the SU(n) group. For $n > 2$, there are no linear conditions which are satisfied by the components of U. We therefore treat the n^2 complex components of U as $2n^2$ independent real variables, on which the constraints $\det U = 1$ and $UU^\dagger = \mathbb{1}$ must be imposed, where $\mathbb{1}$ is the identity matrix. These constraints are not independent because $UU^\dagger = \mathbb{1}$ implies $|\det U| =$

1, so there are n^2+1 independent real constraints in all. A convenient way to express them is

$$\text{Re det } U = 1, \quad \text{Im det } U = 0, \tag{G.1}$$

$$(UU^\dagger)_{\text{traceless}} = 0. \tag{G.2}$$

The last equation implies that U satisfies $UU^\dagger = p^2 \mathbb{1}$, where p is a positive number, and consequently that U is of the form $U = pu$, where u is a unitary matrix, so $|\det u| = 1$. The first pair of equations imply $p^n \det u = 1$, which gives $p = 1$ and $\det u = 1$. Thus these constraints are satisfied by, and only by, elements of the SU(n) group. Moreover, they are regular in the $2n^2$-dimensional real space. For let $U = \mathbb{1} + X$. The conditions are satisfied for $X = 0$, and to first order in X they read tr $X = 0$ and $(X + X^\dagger)_{\text{traceless}} = 0$, so X is traceless and anti-hermitian.

Consider the measure

$$\int \mathrm{d}b \; \delta(\text{Re det } b - 1)\delta(\text{Im det } b)\delta\left[(bb^\dagger)_{\text{traceless}}\right], \tag{G.3}$$

where $\mathrm{d}b$ represents the $2n^2$-dimensional cartesian measure and the last delta-function is of dimension $n^2 - 1$. It is invariant under the change of variables $b' = gb$ and $b'' = bg$, where g is an arbitrary element of the SU(n) group, that correspond to left and right translation on the group. Consequently (G.3) correctly represents the Haar measure for the SU(n) group. We express the delta-functions by integration over Lagrange-multiplier variables, so the Haar measure becomes

$$\int \mathrm{d}b \; \mathrm{d}\mu \; \mathrm{d}\nu \; \mathrm{d}\kappa \; \exp\{-\mu(\text{Re det } b - 1) - \nu(\text{Im det } b) - \text{tr}(\kappa b b^\dagger)\}. \tag{G.4}$$

Here μ and ν are integrated along the imaginary axis, κ is a traceless anti-hermitian matrix (i times a traceless hermitian matrix), and $\mathrm{d}\kappa$ represents integration over its $n^2 - 1$ real components.

This representation of the Haar measure is introduced for each link (xy) in such a way that all linear equations for the components of b_{xy} are preserved. Thus to each link (xy) is assigned a non-unitary matrix b_{xy}, with $b_{yx} = b_{xy}^\dagger \neq b_{xy}^{-1}$. The two color-neutral Lagrange-multiplier variables may be regarded geometrically as link variables that satisfy $\mu_{xy} = \mu_{yx}$ and $\nu_{xy} = -\nu_{yx}$. However, the expression $\text{tr}(\kappa_{xy} b_{xy} b_{xy}^\dagger) = \text{tr}(b_{yx} \kappa_{xy} b_{xy})$ shows that the traceless anti-hermitian matrix κ_{xy} transforms according to $\kappa_{xy} \to g_x^\dagger \kappa_{xy} g_x$ under a gauge transformation. Thus κ_{xy} is a variable associated with the site x, defined by choosing a preferred site at one end of each link, that transforms according to the adjoint representation at site x.

For each link the exponent (G.4) gets added to the action S_0, which in total gets modified by the addition of

$$S_{cs} \equiv \sum_{x,\sigma} \Big(\mu_\sigma(x) \{ \mathrm{Re}\ \det[b_\sigma(x)] - 1 \} + \nu_\sigma(x)\ \mathrm{Im}\ \det[b_\sigma(x)]$$

$$+ \mathrm{tr}[\kappa_\sigma(x) b_\sigma(x) b_\sigma^\dagger(x)] \Big). \tag{G.5}$$

The action is invariant under the BRS transformation s,

$$sS_{cs} = 0, \tag{G.6}$$

where s acts on $b_\sigma(x)$ according to eq. (9.9a), and on the Lagrange-multiplier fields according to

$$s\mu_\sigma(x) = 0, \quad s\nu_\sigma(x) = 0, \tag{G.7}$$

$$s\kappa_\sigma(x) = -C(x)\kappa_\sigma(x) + \kappa_\sigma(x)C(x). \tag{G.8}$$

Sources must be introduced for the $2n^2$ real components of $b_\sigma(x)$, for the $2n^2$ real components of $sb_\sigma(x)$, for $\mu_\sigma(x)$ and $\nu_\sigma(x)$, for the n^2-1 real components of $\kappa_\sigma(x)$, and for the n^2-1 real components of $s\kappa_\sigma(x)$. The remainder of the discussion then follows as in sect. 9.

Appendix H

LATTICE WEAK-COUPLING PERTURBATION THEORY

In this appendix we briefly develop the lattice weak-coupling perturbation theory based on the lattice effective action formalism of sect. 9.

In zeroth order, we have

$$\Gamma^{(0)}(\Phi, K) = S(\Phi, K), \quad \Gamma^{(0)}(\Phi) \equiv \Gamma^{(0)}(\Phi, 0) = S_{phys}(\Phi). \tag{H.1}$$

where S and S_{phys} are given in eqs. (9.13) and (9.12). The perturbation series is obtained from expanding about the approximate stationary point $\Phi^{(0)}$ determined by

$$\partial S_{phys}(\Phi^{(0)})/\partial\Phi_i = 0. \tag{H.2}$$

In the case of the SU(2) group, these equations give, for the four gauge field components

$$b_\sigma^{(0)a}(x) = \delta^{a,0} \tag{H.3}$$

for the Lagrange multiplier field

$$\mu_\sigma^{(0)}(x) = (D-1)\beta + \tfrac{3}{2}\alpha, \tag{H.4}$$

and

$$C^{(0)} = C^{(0)*} = \varphi^{(0)} = \varphi^{(0)*} = \omega^{(0)} = \omega^{(0)*} = 0. \tag{H.5}$$

Corresponding to these values, we have in the tree approximation

$$\Gamma^{(0)}(\alpha) = S_{\text{phys}}(\Phi^{(0)}) = -3(D-1)\alpha. \tag{H.6}$$

As foreseen, the horizon condition in the tree approximation $\partial\Gamma^{(0)}(\alpha)/\partial\alpha = 0$, has no solution.

The lattice weak-coupling perturbative expansion is obtained by setting

$$\beta = 2n/g^2, \tag{H.7}$$

and rescaling α according to

$$\alpha = \gamma/g^2. \tag{H.8}$$

An expansion of the fields about $\Phi^{(0)}$ eliminates linear terms. To make the quadratic part of the action of degree g^0, and to keep the BRS transformations regular at $g=0$, new rescaled and shifted fields A, R, y, λ', C' and $C^{*\prime}$ are introduced according to

$$b_\sigma^0(x) \equiv 1 + gR_\sigma(x), \tag{H.9}$$

$$b_\sigma^a(x) \equiv \tfrac{1}{2}gA_\sigma^a(x) \quad \text{for } a = 1, 2, 3, \tag{H.10}$$

$$\mu_\sigma(x) \equiv (D-1)\beta + \tfrac{3}{2}\alpha + g^{-1}y_\sigma(x), \tag{H.11}$$

$$\lambda^a(x) \equiv g^{-1}\lambda'^a(x), \tag{H.12}$$

$$C^a(x) \equiv gC'^a(x), \tag{H.13}$$

$$C^{*a}(x) \equiv g^{-1}C^{*\prime a}(x), \tag{H.14}$$

the remaining fields being unchanged. The cubic and quartic pieces of S_{phys} contain factors of g and g^2, and the lattice weak-coupling expansion may be effected straightforwardly.

In the naive continuum limit, in which a is the lattice spacing, the field R is suppressed by a mass term of order a^{-2}, and the fields R and y decouple from the remaining fields. This is not strictly true, because the naive continuum limit is ultraviolet divergent and requires dimensional regularization. In a more careful treatment, these fields provide needed renormalization counterterms in the continuum limit of the lattice gauge perturbation theory.

References

[1] I.M. Singer, Commun. Math. Phys. 60 (1978) 7
[2] Gribov, Nucl. Phys. B139 (1978) 1
[3] Semyonov-Tian-Shansky and Franke, Proc. Seminars of the Leningrad Math. Inst., 1982 (Plenum, New York, 1986) [English translation]
[4] P. van Baal, Nucl. Phys. B369 (1992) 259
[5] D. Zwanziger, Nucl. Phys. B378 (1992) 525
[6] D. Zwanziger, Nucl. Phys. B364 (1991) 127
[7] D. Zwanziger, Nucl. Phys. B323 (1989) 513
[8] D. Zwanziger, Nucl. Phys. B399 (1993) 477
[9] H. Neuberger, Phys. Lett. B175 (1986) 69
[10] F. Nill, MPI-PAE/PTh 39/85 (Munich 1985);
H. Neuberger, Phys. Lett. B183 (1987) 337
[11] E. Kraus and K. Sibold, Nucl. Phys. B372 (1992) 113
[12] C.M. Becchi and F. Palumbo, Phys. Rev. D44 (1991) R946
[13] C. Bernard, C. Parrinello and A. Soni, A lattice study of the gluon propagator in momentum space, Washington University preprint HEP/93-33
[14] T. Kugo and I. Ojima, Prog. Theor. Phys. (Suppl.) 66 (1979) 1, eqs. (6.14) and (6.37)